

Hydrogeological Atlas of Rajasthan

Outside Basin



2013

Message



Shri Ashok Gehlot,
Hon'ble Chief Minister, Rajasthan

Ground water is an inseparable aspect in the lives of the people of Rajasthan. There are no major or perennial rivers flowing through the State. Moreover, most part of the state receives very less rainfall. Our dependence on ground water is extremely critical not only as drinking water for human beings and cattle but also as major source of irrigation.

The Government of Rajasthan is keen on sustainable utilization of the ground water. The State water policy, adopted in 2010 emphasizes on community participation in ground water management and use of information technology for widespread dissemination of data for the benefit of all sections of the society.

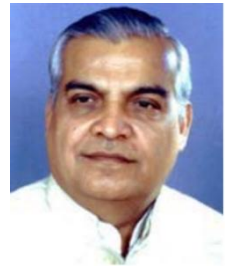
I am pleased to note that the Ground Water Department of the State has carried a detailed assessment of quality, quantity and distribution of ground water resources of the State through mapping of aquifers and using most modern tools and techniques.

It is also heartening to note that the outcome of the project is being published in the form of '*Hydrogeological Atlas of Rajasthan*'. It will incorporate the scientific findings and reliable information in various types of readily understandable GIS maps based on ground water potential viz. zone maps, depth to water level maps, ground water categorization maps, chemical quality maps, aquifer maps etc. These would be certainly found useful by experts and user agencies. The same would also help in raising the awareness levels of the various stakeholders in utilizing the precious ground water resources of the State.

I take this opportunity to congratulate the Ground Water Department officials for bringing this useful publication and wish them success in all such endeavor.

(Ashok Gehlot)

Message



Dr. Jitendra Singh,
Hon'ble Minister for Energy,
Non-Conventional Energy Resources,
PHED, GWD, Information & Public Relations

Ground water is a renewable resource but its long term sustainability depends on the balance of recharge and withdrawal ground water is added every year through rainfall and other sources. In Rajasthan, volume of water extracted for different purposes far exceeds its replenishment which has led to depletion of ground water in large parts of state.

While the surface water is visible and largely measurable, the ground water is hidden underneath. To meet the ever increasing demand of water, the state has been promoting conjunctive use of water. In order to have a state level estimation of water resources, the estimation of ground water quality and quantity is a key input. The studies being very scientific in nature it remains a challenge how to communicate the message to all stakeholders.

The publication of '*Hydrogeological Atlas of Rajasthan*' is a major step forward and will help in visualization of ground water related information in very simplified graphic and tabular forms. The effort that has gone into creation of the atlas from basic information in archives is very much appreciated.

(Jitendra Singh)

Message



Dr. Purushottam Agarwal

Principal Secretary to Government,
Public Health Engineering and Ground Water Department, Rajasthan

Ground water has been the principal source over years to meet the needs of human and cattle population of the state. Increasingly modern techniques are used for extraction of large volume of ground water which has exceeded recharge in most parts of the State. As a result, water table in most of the areas has gone down significantly. People are digging deeper and deeper to reach the ground water to meet their increasing needs. Maintaining a healthy practice of balance between recharge and withdrawal is critical for its sustainability.

The state had initiated various programs under which a systematic assessment of water resources is being carried out and all future water management plans are chalked out on the basis of sound database and historic data analysis. Increasing population, industrialization, agricultural need, power generation etc. are bound to pose challenges before us in coming years and to face them we have documented 'Water Resource Vision 2045' which highlights the short term (upto 2015) and long term (upto 2045) thrust areas and action plan which are pre-requisites for successful implementation of the State Water Policy. We are committed to achieving the objective of optimum use of every drop of scarce and precious utilizable water resource. In this regard, the Ground Water Department has also carried out a systematic assessment of ground water resources and three dimensional mapping of aquifers that gives a new insight.

I congratulate the Ground Water Department on publication of '*Hydrogeological Atlas of Rajasthan*' which emphasizes on various aspects of ground water which I feel will help us in spreading the information on current scenario and consequently help in managing it better.

(Purushottam Agarwal)

Foreword



Arno Schaefer

Minister Counselor, Head of Operations

The Government of Rajasthan has entered into an agreement with the European Union through the State Partnership Programme for implementing state-wide water sector reform leading to sustainable integrated water resources management in Rajasthan, and supporting the Panchayat Raj institutions in executing their responsibilities in equitable access to safe, adequate, affordable and sustainable water supply as well as conservation, stabilization and replenishment of surface and ground water.

Ground water is the primary source for drinking, domestic, agriculture and industries and other purposes in Rajasthan. The stage of ground water development is close to 135% i.e., if the net annual ground availability in Rajasthan is about 10.7 billion cubic meter, the annual draft is 14.5 billion cubic meter. As a result of this, the water levels have been depleting alarmingly. The ground water in 166 out of 249 blocks of the state fall under the category of over-exploited. This indiscriminate and sometime exploitative use of ground water has led to questions regarding its sustainability.

This publication '*Hydrogeological Atlas of Rajasthan*' is the culmination of a long term effort of collation, integration, interpretation and analysis of vast archive of historic data obtained by Rajasthan Ground Water Department. The atlas is in three parts, (1) Statewide Atlas, which shows the state of ground water, water quality, current stage of exploitation, aquifer mapping and basic information on administrative setup, climate, geology and geomorphology; (2) A detailed series, wherein compilation of information and district level thematic maps and all 33 districts have been separately presented and (3) Compilation of information at the River Basin level. In addition to the thematic maps, a large number of cross sections and the 3 dimensional features of aquifers have also been presented.

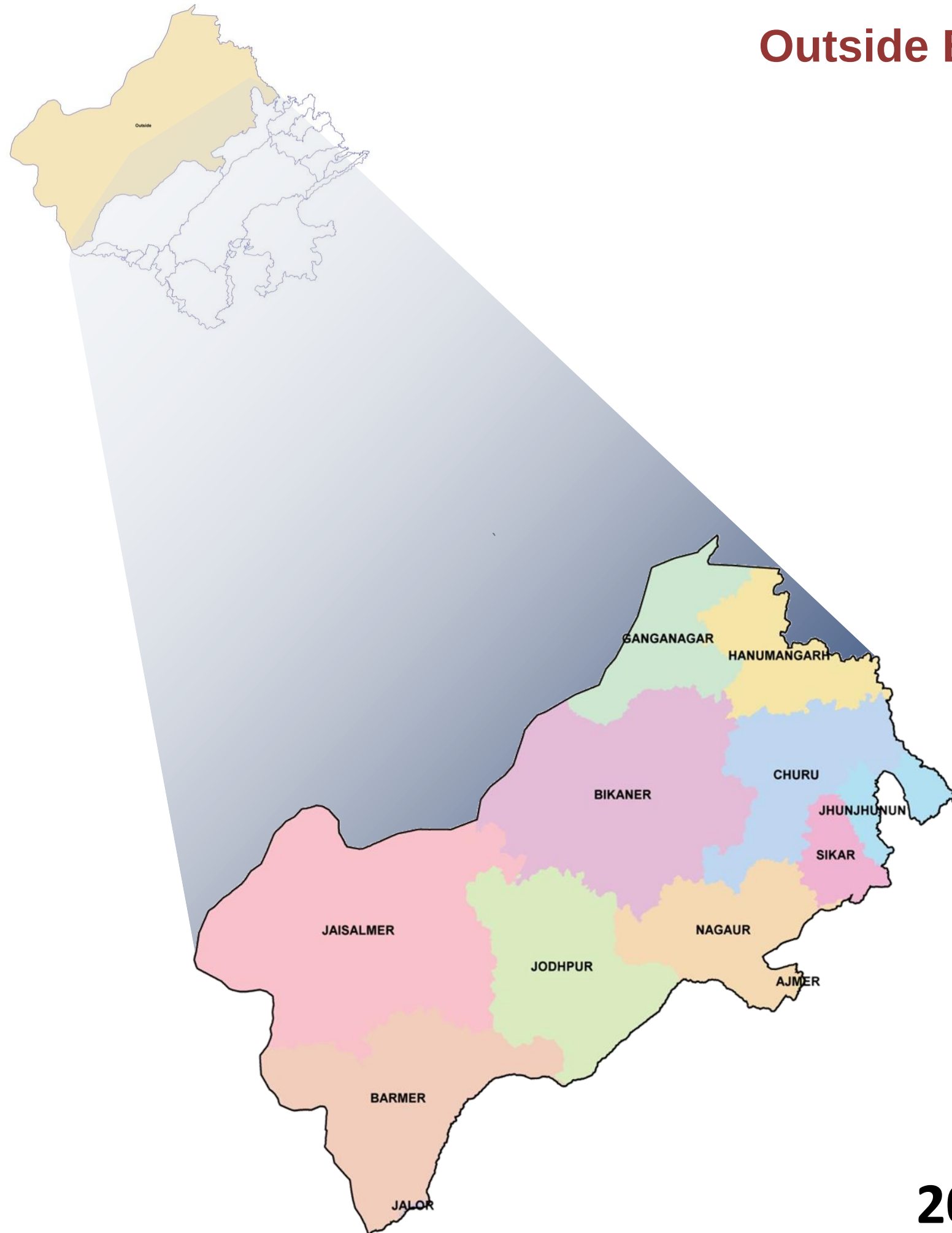
I am convinced that the Atlas will contribute to better understand ground water and help managerial and future planning. The current Geographic Information System (GIS) based study is a pioneering step in India and will pave the way for similar approach to be adopted by other Indian States.

I appreciate very much the efforts made by the Ground Water Department, Government of Rajasthan, The European Union Technical Assistance Team and M/s Rolta India Limited for undertaking this study and compiling this Atlas under the India-EU State partnership Programme.

(Arno Schaefer)

Hydrogeological Atlas of Rajasthan

Outside Basin



Contents:

List of Plates	Title	Page No.
Plate I	Administrative Map	2
Plate II	Topography	4
Plate III	Rainfall Distribution	4
Plate IV	Geological Map	6
Plate V	Geomorphological Map	6
Plate VI	Aquifer Map	8
Plate VII	Location of Ground Water Monitoring Stations	8
Plate VIII	Location of Exploratory Wells	10
Plate IX	Depth to Water Level (Pre-Monsoon 2010)	10
Plate X	Water Table Elevation (Pre-Monsoon 2010)	12
Plate XI	Water Level Fluctuation (Pre-Post Monsoon 2010)	12
Plate XII	Electrical Conductivity Distribution (Average Pre-Monsoon 2005-09)	14
Plate XIII	Chloride Distribution (Average Pre-Monsoon 2005-09)	14
Plate XIV	Fluoride Distribution (Average Pre-Monsoon 2005-09)	16
Plate XV	Nitrate Distribution (Average Pre-Monsoon 2005-09)	16
Plate XVI	Depth to Bedrock	18
Plate XVII	Map of Unconfined Aquifer	18
Plate XVIII	Index Map Showing Alignment of Cross Sections	20
Plate XIX	Cross Section Along A-A'	22
Plate XX	Cross Section Along B-B'	22
Plate XXI	Cross Section Along C-C'	24
Plate XXII	Cross Section Along D-D'	24
Plate XXIII	Cross Section Along E-E'	26
Plate XXIV	3D Model	28
Glossary of terms		29

2013

ADMINISTRATIVE SETUP

Location:

The Outside Basin is not a river basin, but it is the vast semi-desert area apart of the Great Indian Thar Desert which extends over almost the entire western drainage of Rajasthan, gently sloping from the Aravali range to the India-Pakistan border; it lacks any distinct surface water catchment features. The Outside basin lies in the western and northern parts of Rajasthan, It extends between 24° 41' 19.69" to 30° 12' 08.60" North latitude and 69° 18' 41.91" to 76° 07' 07.12" East Longitudes as shown in Plate – I encompassing an area of approximately 1,65,842 km² (including Ghaggar sub basin). The international boundary with Pakistan constitutes the western limit of the basin while Punjab and Haryana form the northern boundaries. In the east and southeast, the boundary is defined by the Shekawati and Luni River Basins. The basin roughly 600km long, from northeast to southwest and about 280 km wide, from the Aravalis to the border with Pakistan.

Administrative Set-up:

The Outside Basin extends over parts of districts Ajmer, Barmer, Bikaner, Churu, Ganganagar, Hanumangarh, Jaisalmer, Jalor, Jhunjhunun, Jodhpur, Nagaur and Sikar divided into 98 Blocks encompassing 11,898 villages.

S. No.	District Name	Area (sq km)	% of Basin Area	Total Number of Blocks	Total Number of Towns and Villages
1	Ajmer	48.7	-	3	6
2	Barmer	21,927.7	13.2	8	1,521
3	Bikaner	30,279.3	18.3	11	879
4	Churu	13,817.3	8.3	6	888
5	Ganganagar	10,684.0	6.3	19	3,026
6	Hanumangarh	9,929.2	6.0	13	1,911
7	Jaisalmer	38,487.2	23.2	3	639
8	Jalor	92.0	0.1	1	5
9	Jhunjhunun	3,597.8	2.2	8	573
10	Jodhpur	19,200.3	11.6	9	861
11	Nagaur	13,357.0	8.1	11	1,049
12	Sikar	4,421.6	2.7	6	540
Total		1,65,842.1	100.0	98	11,898

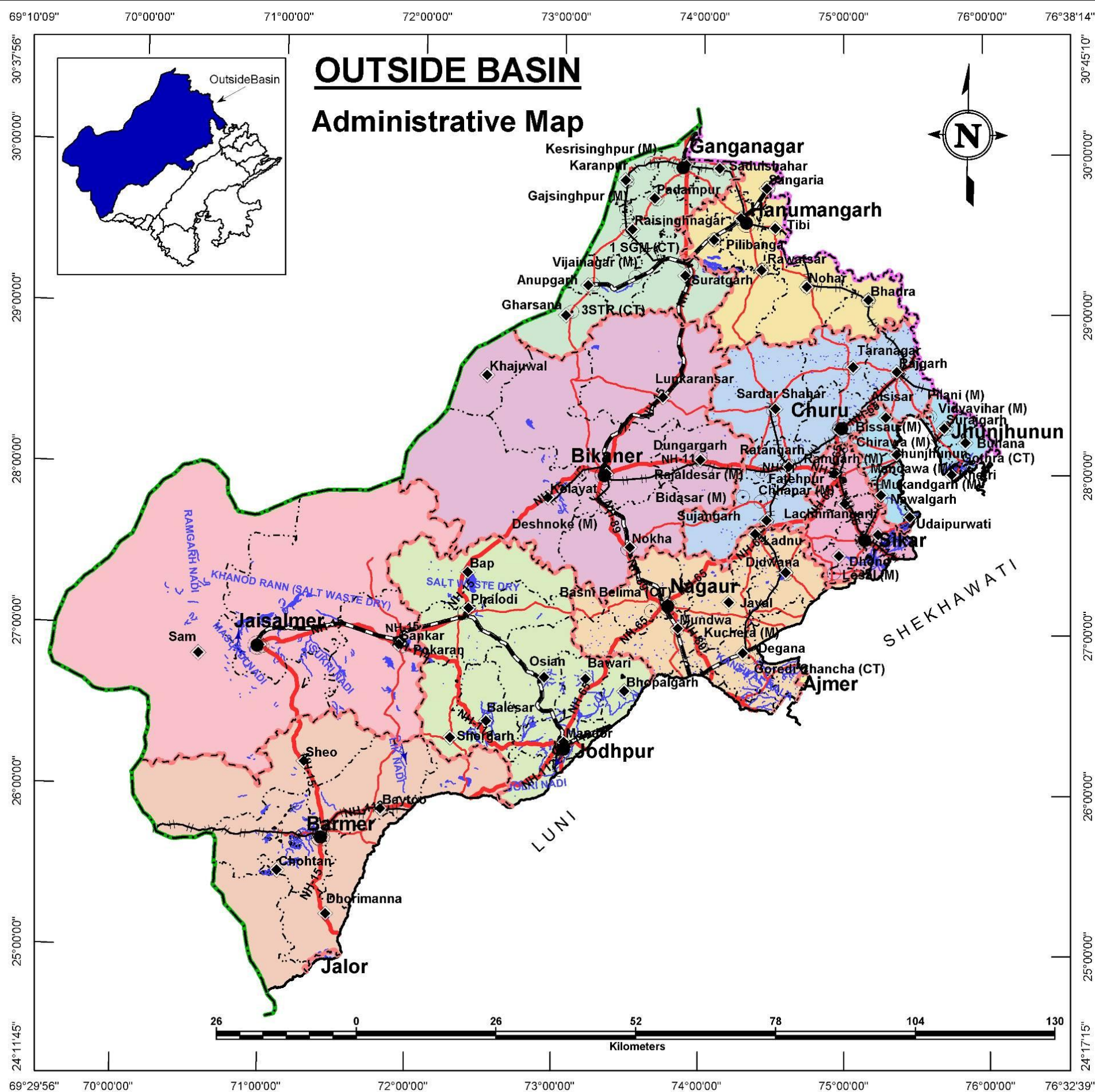
Climate:

The basin is very large, covering almost half of Rajasthan in its western part but it falls into Arid climatic zone which is typically exemplified by dry conditions throughout the year and temperature extremes of very high (reaching upto 50°C) in mid-summer to very low (down to 1 to 2°C) in winter nights. Even the diurnal temperature variations are very high with hot days and cold nights for the most part of the year. Rainfall is scanty dipping to the lowest mean annual rainfall of about 100mm in the desert areas to high of upto 400mm in eastern parts adjoining Aravali hills. With almost negligible monsoon season, the two pronounced seasons are summer and winter.

OUTSIDE BASIN



PLATE - I



LEGEND

Admin Boundary:

District Headquarter



Block Headquarter



Town



International Boundary



State Boundary



District Boundary



Block Boundary



Basin Boundary



Roads:

National Highway



State Highway



Railway:

Broad Gauge



Metre Gauge



Water Bodies:

River / Streams



Ponds / Reservoirs



Hills



TOPOGRAPHY

Most of the basin area is represented by low undulating stabilized and active dunal plains but bordering areas in the southernmost part of Barmer district has minimum elevation of 14m amsl and the hills in Jhunjhun district have maximum elevation of about 1037 m amsl.

Table: District wise minimum and maximum elevation

S. No.	District Name	Min Elevation (m amsl)	Max. Elevation (m amsl)
1	Ajmer	436.2	567.2
2	Barmer	14.1	622.6
3	Bikaner	105.6	348.2
4	Churu	192.6	478.0
5	Ganganagar	104.4	230.0
6	Hanumangarh	164.2	240.7
7	Jaisalmer	31.5	347.8
8	Jalor	16.2	90.4
9	Jhunjhun	259.6	1,036.8
10	Jodhpur	145.1	445.3
11	Nagaur	273.9	726.1
12	Sikar	294.5	1,026.7

OUTSIDE BASIN

RAINFALL

Table: District wise total annual rainfall (based on year 2010 meteorological station recordings (<http://waterresources.rajasthan.gov.in>))

S. No.	Rain gauge Stations	Total Monsoon Rainfall (mm)	Total Non-Monsoon Rainfall (mm)	Total Annual Rainfall (mm)
1	Anupgarh	251	21	272
2	Baitu	372	50	422
3	Barmer	340	94	434
4	Bhadra	594	44	638
5	Bhopalgarh	462	32	494
6	Bikaner	379	30	409
7	Buhana	517	61	578
8	Chirawa	494	72	566
9	Chohtan	544	16	560
10	Churu	570	65	635
11	Degana	541	64	605
12	Didwana	592	10	602
13	Fatehgarh	362	34	396
14	Fatehpur	629	44	673
15	Ganganagar	351	15	366
16	Hanumangarh	519	28	547
17	Jaisalmer	285	24	309
18	Jayal	441	28	469
19	Jhunjhunu	775	53	828
20	Karanpur	282	31	313
21	Khetri	895	49	944

S. No.	Rain gauge Stations	Total Monsoon Rainfall (mm)	Total Non-Monsoon Rainfall (mm)	Total Annual Rainfall (mm)
22	Ladnu	458	30	488
23	Lunkaransar	767	16	783
24	Malsisar	932	59	991
25	Nagaur	406	25	431
26	Nawalgarh	575	80	655
27	Nohar	468	34	502
28	Osian	411	24	435
29	Padampur	457	23	480
30	Pokaran	520	20	540
31	Raisingh Nagar	328	8	336
32	Ratangarh	666	47	713
33	Sadulshahar	286	36	322
34	Sam	225	2	227
35	Sardarshahar	536	20	556
36	Sheo	401	27	428
37	Sikar	818	78	896
38	Sujargarh	552	75	627
39	Suratgarh	406	28	434
40	Taranagar	780	22	802
41	Udaipurwati	748	65	813

GEOLOGY

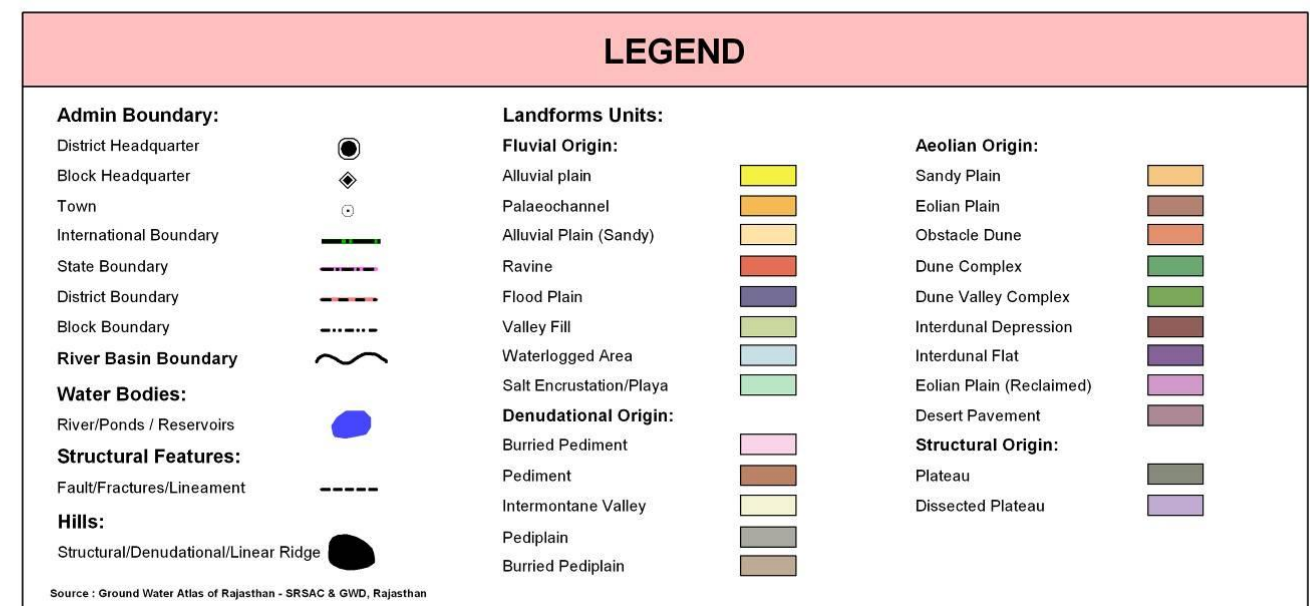
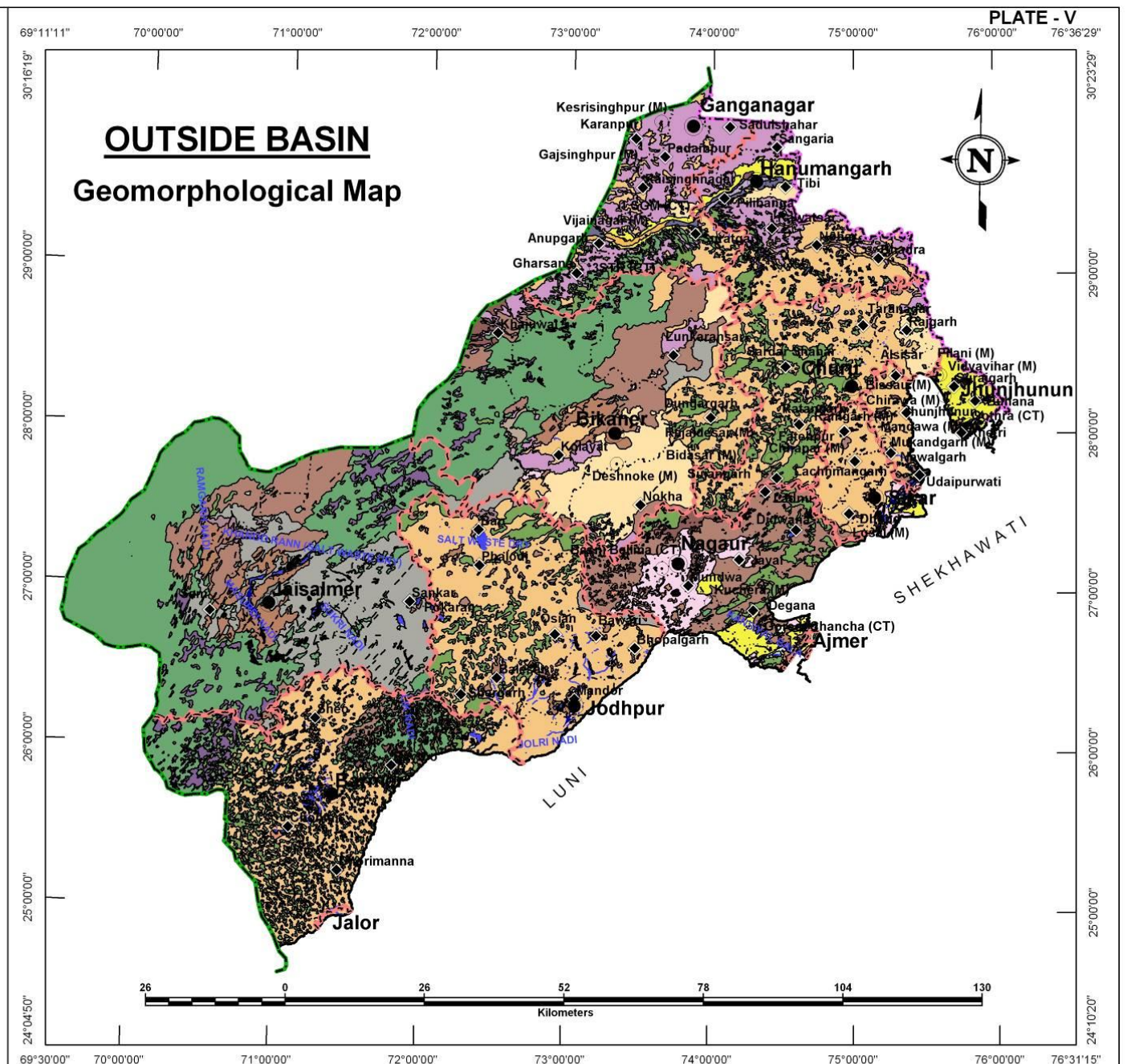
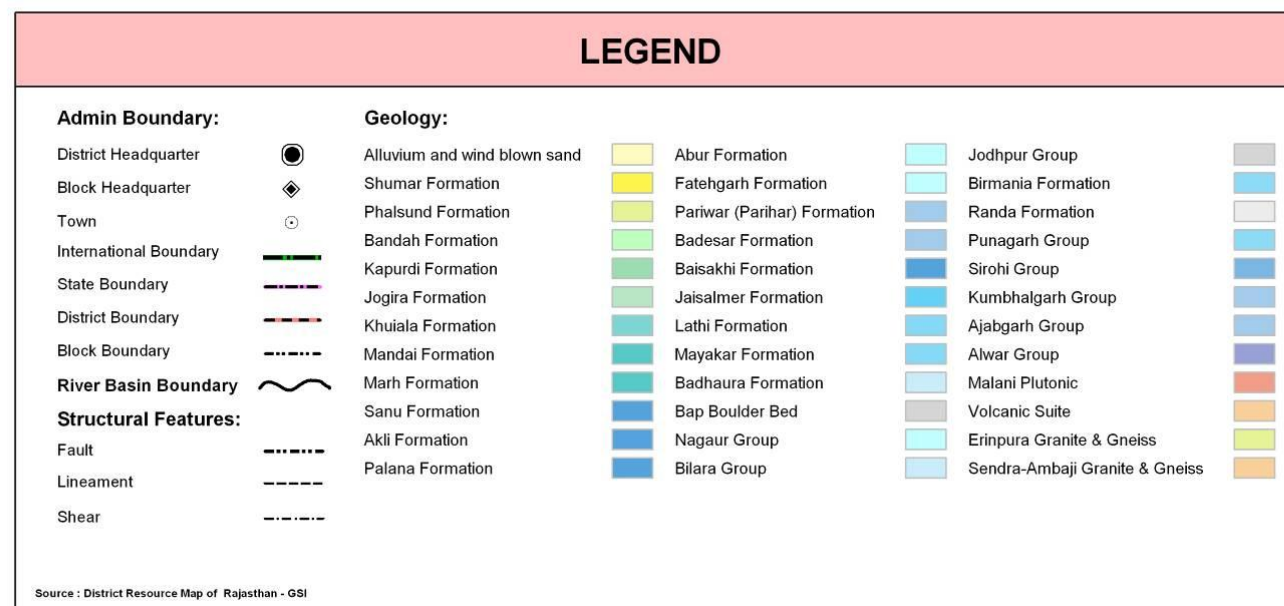
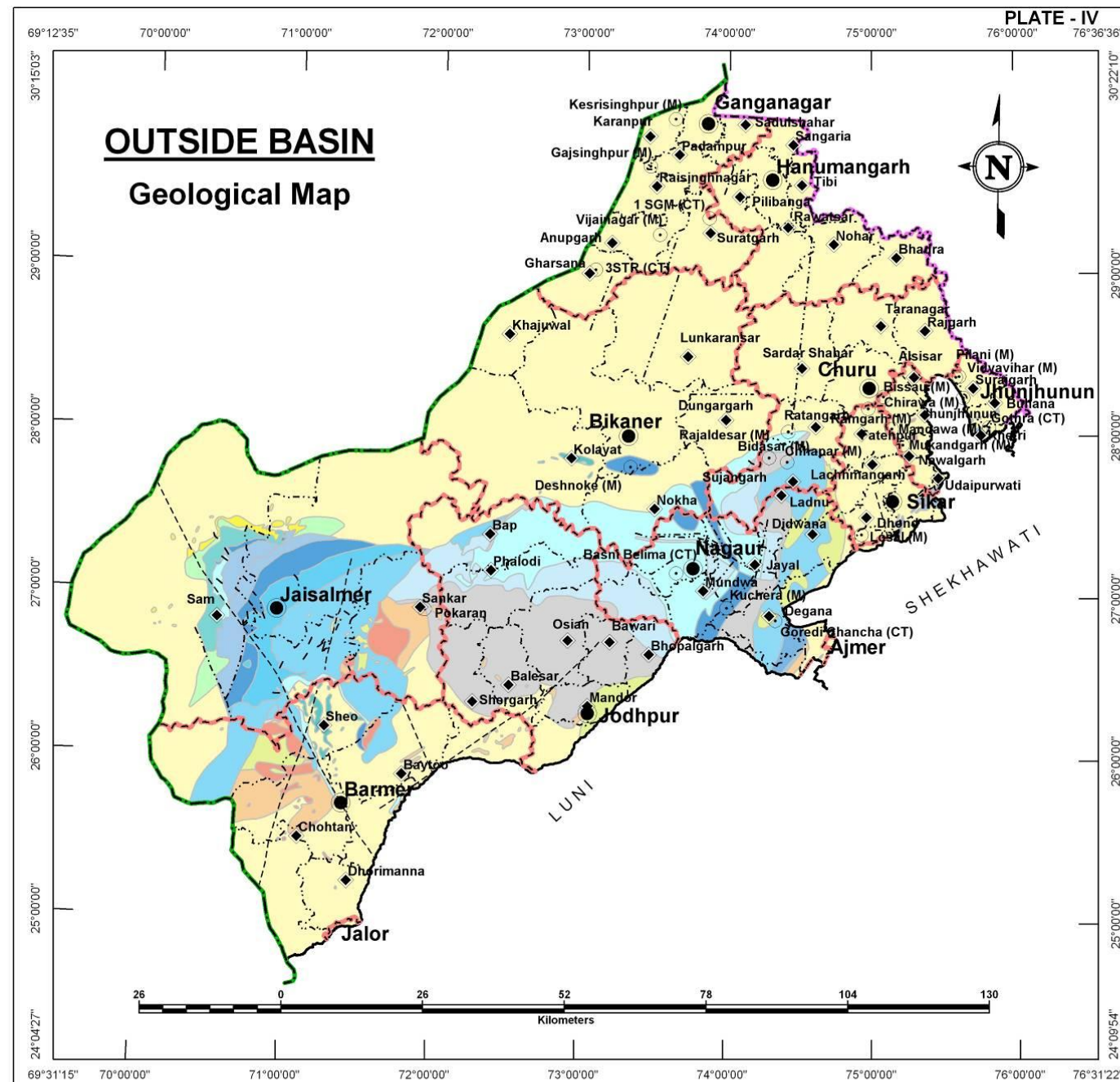
Owing to vast spatial extent of the basin rocks of wide ranging ages are found from Archean to Recent age and well represented by rocks of Pre-Cambrian, Permo-Carboniferous, Jurassic, Cretaceous, Paleocene and Recent age.

OUTSIDE BASIN

Age/Super-Group	Group/Formation	Rock Types
Recent to Sub-Recent	Alluvium & Sand dunes	Sand, silt & clay
	-----X-----X-----X-----X-----Unconformity-----X-----X-----X-----	
Palaeocene to Eocene	Banda/Khuiala/Sanu/	Sandstone,
	Palana Series	Limestone, bentonitic clay, Fuller's earth and lignite seams
	-----X-----X-----X-----X-----Unconformity-----X-----X-----X-----	
Upper Cretaceous to Palaeocene	(Deccan Trap)	Basaltic lava flows with inter-Trappean sediments
	Abur Series	Sandstone, limestone, clay and lignite
Cretaceous	Parihar Series	Felspathic sandstone
	Badesar Series	Ferruginous sandstone
	Baisakhi Series	Calcareous sandstone
Jurassic	Jaisalmer Series	Fossiliferous limestone, sandstone, etc.
	Lathi Series	Sandstone, shale, etc.
	-----X-----X-----X-----X-----Unconformity-----X-----X-----X-----	
Permo Carboniferous	Badhaura Series (Bap Boulder Bed)	Sandstone and boulders
	-----X-----X-----X-----X-----Unconformity-----X-----X-----X-----	
Lower to Upper Cambrian (?) (Marwar Super-Group)	Nagaur Group	Sandstone, limestone, gypsum and siltstone,
	Bilara Group	Limestone and dolomite
	Jodhpur Group	Sandstone, shale and boulder bed
	-----X-----X-----X-----X-----Unconformity-----X-----X-----X-----	
Lower to Upper Precambrian (post-Delhi Intrusives)	Malani Rhyolite, Jalore, Erinpura Granites, Siwana	Acid, basic and ultrabasic intrusives and extrusives
	-----X-----X-----X-----X-----Unconformity-----X-----X-----X-----	
Lower to Upper Precambrian (Delhi Super-Group)	Ajabgarh Group (Rhythmites, Pilani Formation)	Quartzite and mica-schist, calc-schist and gneiss, marble, basic flows etc.
	Alwar Group	Felspathic Quartzite, with intercalations of magnetite, hematite bands conglomerate and mica schist
	Sirohi Group	Phyllite, mica schist and biotite schist
	Punagarh Group	Shale, slate, phyllite, mica schist and quartzite
	Raialo Group	Marble (Makrana Quarries), dolomite. Quartzite, schists and conglomerates
	(Granite and basic intrusives)-----	
Bhilwara Super-Group	Ranthambhor Group	Phyllite, slate, schist, gneiss granite, migmatite, etc.

GEOMORPHOLOGY

Origin	Landform Unit	Description
Aeolian	Desert Pavement	Flat barren stony desert plain having coarse, angular, wind polished gravel and small stones lying on rock cut surface.
	Dune Complex	An undulating plain composed of number of sand dunes of crescent shape.
	Dune Valley Complex	Cluster of dunes and interdunal spaces with undulating topography formed due to wind-blown activity, comprising of unconsolidated sand and silt.
	Eolian Plain	Formed by aeolian activity, with sand dunes of varying height, size, slope. Long stretches of sand sheet. Gently sloping flat to undulating plain, comprised of fine to medium grained sand and silt. Also scattered xerophytic vegetation.
	Eolian Plain (Reclaimed)	Gently sloping with sheet of sand or sand dunes, scattered xerophytic vegetation.
	Interdunal Depression	Slightly depressed area in between the dunal complex showing moisture and fine sediments.
	Interdunal Flat	Flat, narrow land between dunes.
	Obstacle Dune	Formed on windward/leeward sides of obstacle like isolated hills or continuous chain of hills, dune to obstruction in path of sand laden winds. Badly dissected well cemented and vegetated.
Denudational	Sandy Plain	Formed of aeolian activity, wind-blown sand with gentle sloping to undulating plain, comprising of coarse sand, fine sand, silt and clay.
	Buried Pediment	Pediment covers essentially with relatively thicker alluvial, colluvial or weathered materials.
	Intermontane Valley	Depression between mountains, generally broad & linear, filled with colluvial deposits.
	Pediment	Broad gently sloping rock flooring, erosional surface of low relief between hill and plain, comprised of varied lithology, criss-crossed by fractures and faults.
Fluvial	Pediplain	Coalescence and extensive occurrence of pediment.
	Alluvial Plain	Mainly undulating landscape formed due to fluvial activity, comprising of gravels, sand, silt and clay. Terrain mainly undulating, produced by extensive deposition of alluvium.
	Alluvial Plain (Sandy)	Flat to gentle undulating plain formed due to fluvial activity, mainly consists of gravels, sand, silt and clay with unconsolidated material of varying lithology, predominantly sand along river.
	Flood Plain	The surface or strip of relatively smooth land adjacent to a river channel formed by river and covered with water when river over flows its bank. Normally subject to periodic flooding.
	Paleochannel	Mainly buried on abandoned stream/river courses, comprising of coarse textured material of variable sizes.
	Valley Fill	Formed by fluvial activity, usually at lower topographic locations, comprising of boulders, cobbles, pebbles, gravels, sand, silt and clay. The unit has consolidated sediment deposits.
	Ravine	Small, narrow, deep, depression, smaller than gorges, larger than gulley, usually carved by running water.
Structural	Salt Encrustation/Playa	Topographical depression comprising of clay, silt, sand and soluble salts, usually undrained and devoid of vegetation.
	Water logged/ Wetland	Area submerged in water or area having very shallow water table. So that it submerges in water during rainy season.
Hills	Dissected Plateau	Plateau, criss-crossed by fractures forming deep valleys.
	Plateau	Formed over varying lithology with extensive, flat, landscapes, bordered by escarpment on all sides. Essentially formed horizontally layered rocky marked by extensive flat top and steep slopes. It may be criss crossed by lineament.
	Denudational, Structural Hill, Linear Ridge	Steep sided, relict hills undergone denudation, comprising of varying lithology with joints, fractures and lineaments. Linear to arcuate hills showing definite trend-lines with varying lithology associated with folding, faulting etc. Long narrow low-lying ridge usually barren, having high run off may form over varying lithology with controlled strike.



AQUIFERS

Alluvium (both younger and older alluvium combined) forms the most predominant aquifer type in the Basin covering almost 50% of the basin area. Next most predominant aquifer type is Sandstone belonging to Nagaur-Jodhpur and Tertiary formations which accounts for about 28% of area. Limestone, Sandstone, Rhyolite, Granite, and Schist together constitute the rest of the aquifer zones however of limited spatial extent.

Aquifer in Potential Zone	Area (sq km)	% of Basin Area	Description of the unit/Occurrence
Younger Alluvium	33,861.4	20.4	It is largely constituted of Aeolian and Fluvial sand, silt, clay, gravel and pebbles in varying proportions
Older Alluvium	48,718.8	29.4	This lithounit comprises of mixture of heterogeneous fine to medium grained sand, silt and kankar
Bilara Limestone	8,243.6	5.0	It is grey to buff coloured hard and compact
Tertiary Sandstone	24,035.7	14.3	Medium to coarse grained, consolidated to semi consolidated sandstone
Nagaur & Jodhpur Sandstone	23,318.3	14.1	Buff to reddish brown in colour, fine to medium grained hard and compact sandstone
Sandstone	13,867.0	8.4	Fine to medium grained, red colour and compact and at places
Parewar Sandstone	1,285.4	0.8	It is feldspathic ferruginous sandstone
Schist	2,873.9	1.7	Medium to fine grained compact rock. The ltho units are soft, friable and have closely spaced cleavage
Rhyolite	6,088.6	3.7	Rhyolite is porphyritic and has phenocryst of quartz and feldspar
Quartzite	596.2	0.4	Medium to coarse grained and varies from feldspathic grit to sericitic quartzite.
Granite Jalore	709.5	0.4	Grey to pink in colour, medium to coarse grained, and non-porphyritic
Granite	1,903.0	1.2	Light grey to pink colour, medium to coarse grained, and characteristically have porphyritic texture
Non Potential Zone	340.7	0.2	Hills
Total	1,65,842.1	100.0	

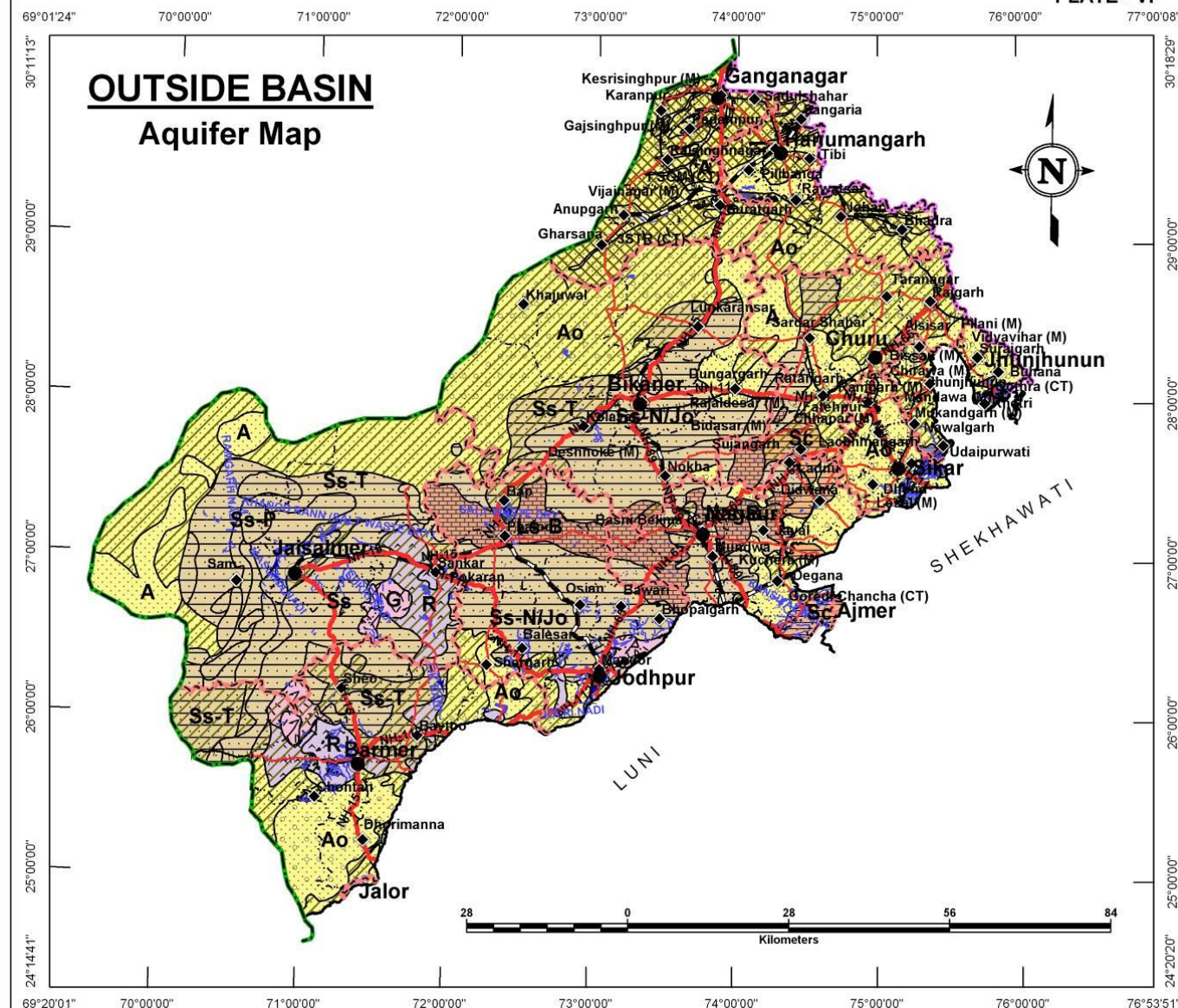
LOCATION OF GROUND WATER MONITORING WELLS

The Outside basin has a large number of monitoring wells however with varying density of distribution over the different parts of the basin. The western desert areas have lower density of monitoring wells whereas the density increases towards east. There are 2,462 ground water monitoring wells in the basin owned by RGWD (1932) and CGWB (530). An additional 605 wells have been recommended to be added to network to more effectively monitor ground water level and quality monitoring of the basin.

District Name	CGWB	RGWD	Total	Recommended additional wells for optimization of monitoring network	
				Water Level	Water Quality
Ajmer		2	2	-	-
Barmer	77	197	274	11	25
Bikaner	106	338	444	-	148
Churu	77	165	242	-	59
Ganganagar	42	127	169	-	11
Hanumangarh	45	113	158	-	30
Jaisalmer	96	223	319	297	13
Jalor	-	-	-	-	-
Jhunjhunun	32	55	87	-	99
Jodhpur	19	386	405	-	29
Nagaur	12	223	235	-	49
Sikar	24	103	127	-	142
Total	530	1,932	2,462	308	605



PLATE - VI



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- International Boundary
- State Boundary
- District Boundary
- Block Boundary
- Basin Boundary

Roads:

- National Highway
- State Highway

Railway:

- Broad Gauge
- Metre Gauge

Water Bodies:

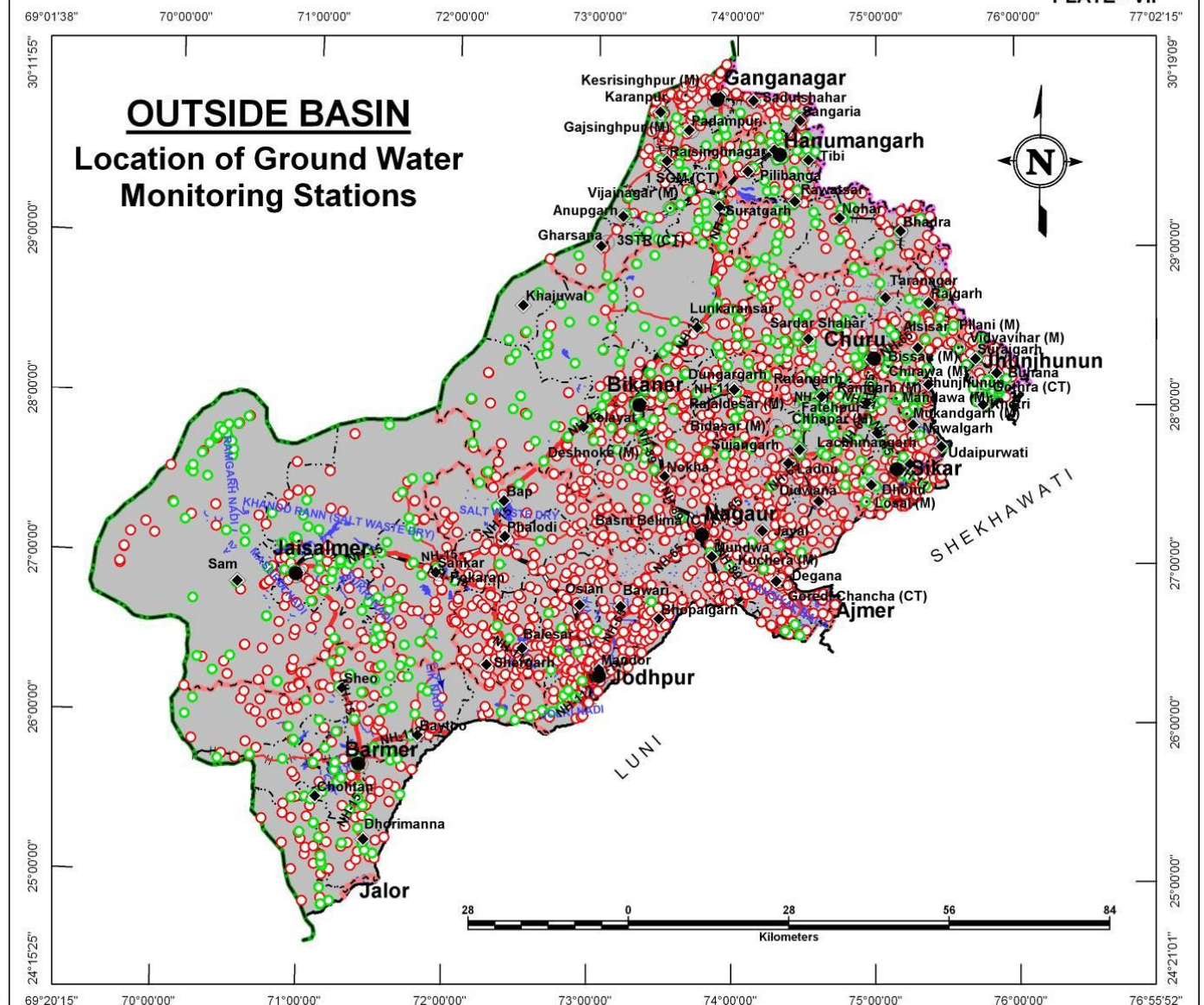
- River / Streams
- Ponds / Reservoirs
- Hills
- Chemically Unsuitable
- Command Area
- Saline Area

Aquifer:

- Younger Alluvium
- Older Alluvium
- Bilara Limestone
- Tertiary Sandstone
- Nagaur & Jodhpur Sandstone
- Sandstone
- Parewar Sandstone
- Schist
- Rhyolite
- Quartzite
- Granite Jalore
- Granite
- Gneiss

Source: Ground Water Potential Zone Map - GWD, Rajasthan

PLATE - VII



LEGEND

Admin Boundary:

- District Headquarter
- Block Headquarter
- Town
- International Boundary
- State Boundary
- District Boundary
- Block Boundary
- Basin Boundary

Roads:

- National Highway
- State Highway

Railway:

- Broad Gauge
- Metre Gauge

Water Bodies:

- River / Streams
- Ponds / Reservoirs
- Hills

Ground Water Monitoring Stations:

- CGWB
- RGWD

LOCATION OF EXPLORATORY WELLS

In all there are 1,432 exploratory wells present in the basin drilled in the past by RGWD (1,054) and CGWB (378) that are well distributed over the basin area. This constitutes the basis for preparation of sub-surface maps and leading to understanding of three dimensional distribution of aquifer systems.

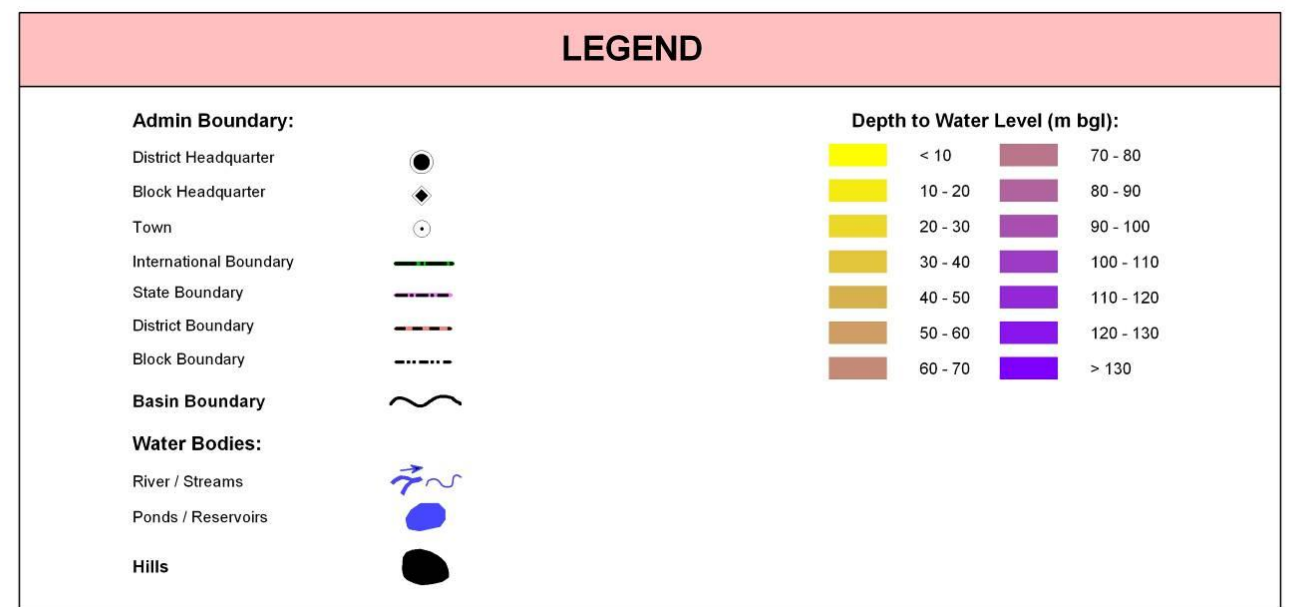
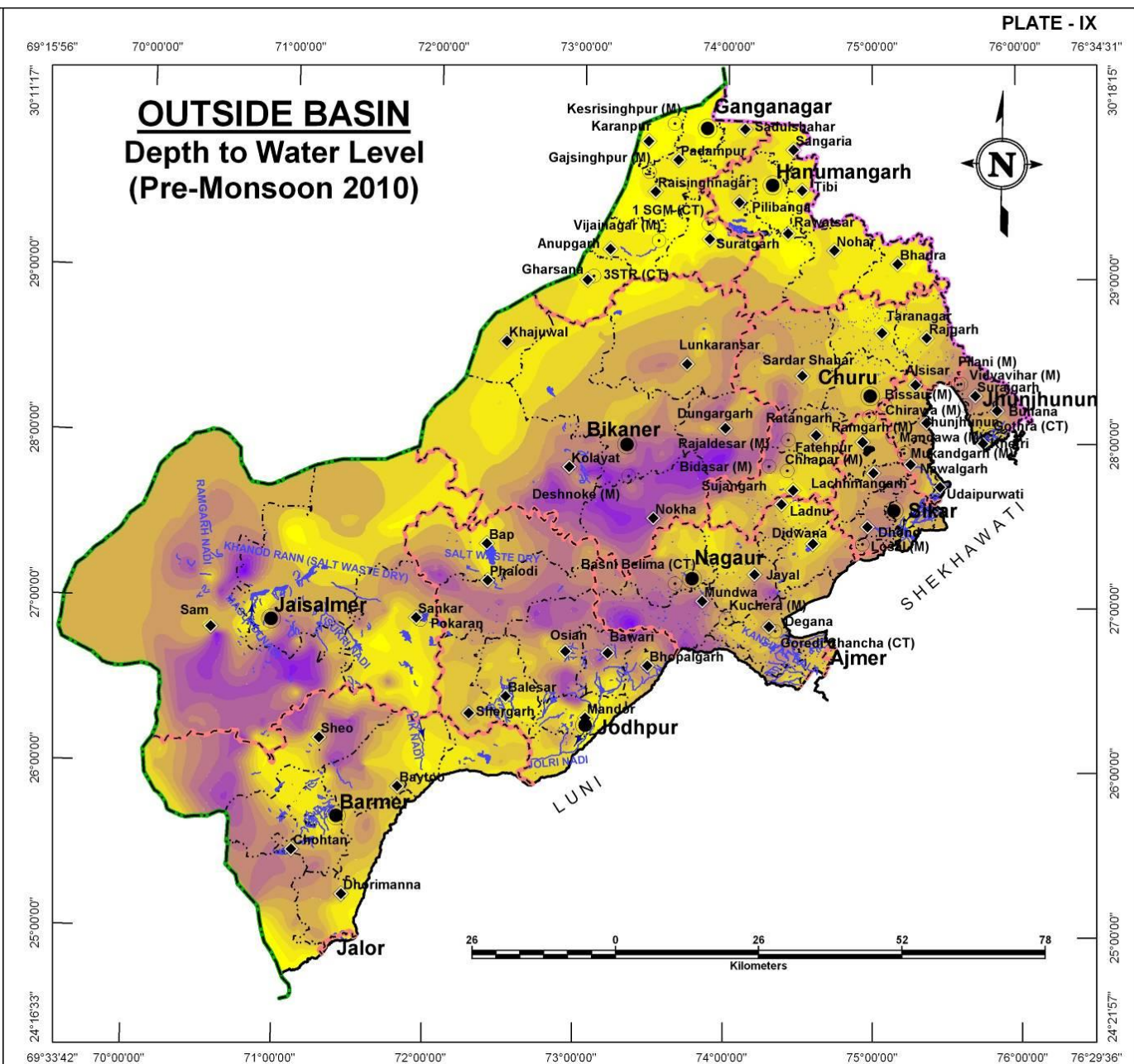
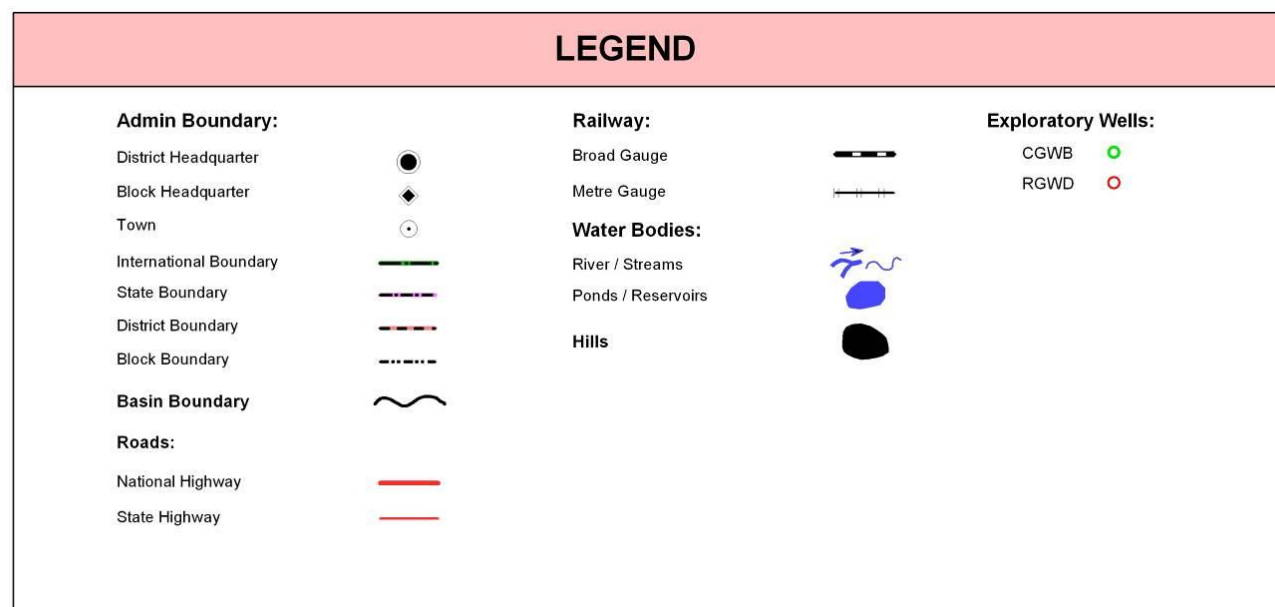
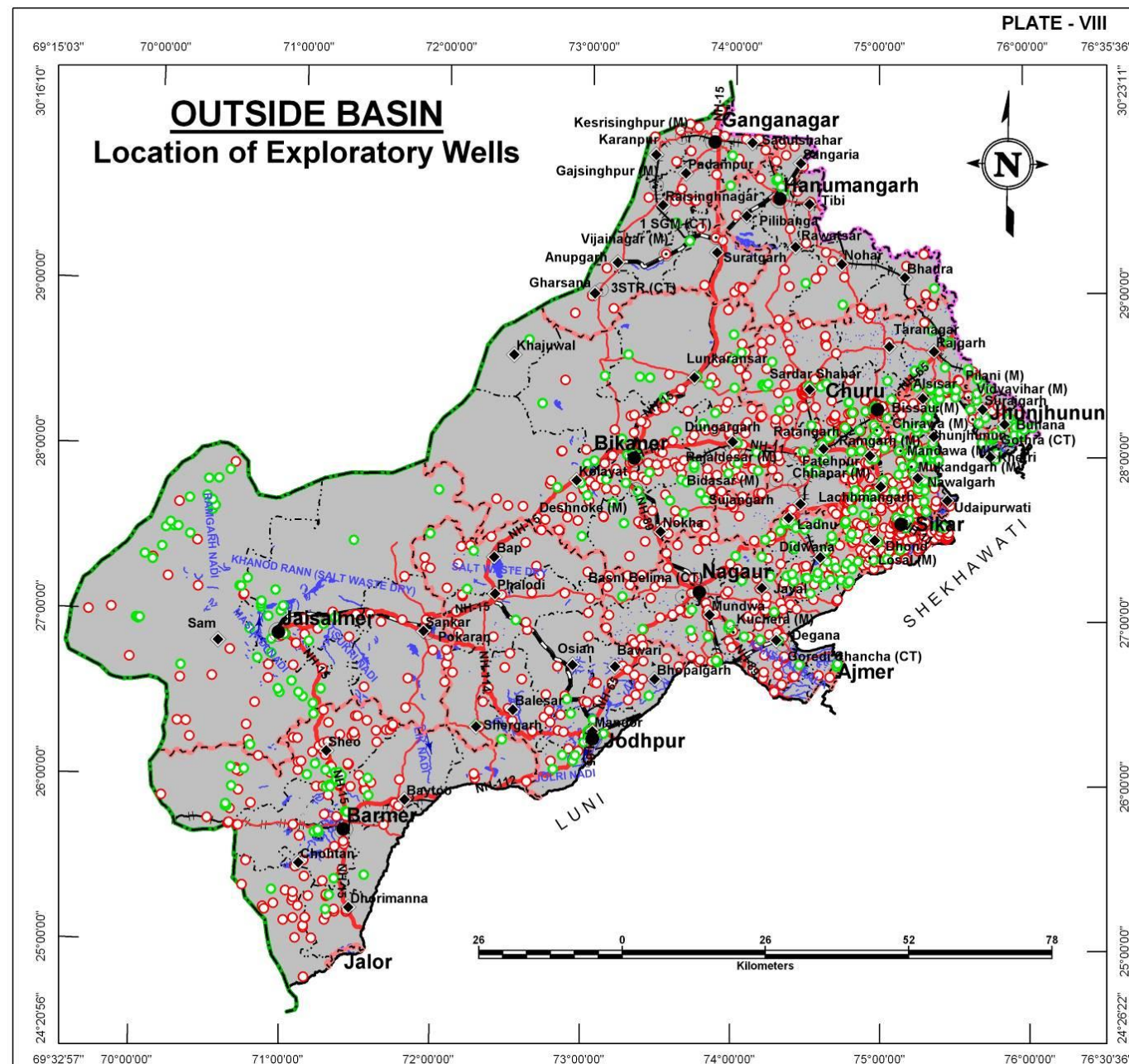
District Name	CGWB	RGWD	Total
Ajmer	2	-	2
Barmer	31	110	141
Bikaner	49	145	194
Churu	60	154	214
Ganganagar	3	33	36
Hanumangarh	8	28	36
Jaisalmer	50	49	99
Jalor	-	-	-
Jhunjhunun	59	52	111
Jodhpur	22	72	94
Nagaur	52	126	178
Sikar	42	285	327
Total	378	1,054	1,432

DEPTH TO WATER LEVEL (PRE MONSOON – 2010)

Depth to water level in the basin shows large variation ranging from less than 10m below ground level in the north around Hanumangarh and Ganganagar to more than 130m bgl to the east of Bikaner. The areas around Jaisalmer also show very deep water levels which varies from 40m bgl to around 130m bgl. Other deep water level pockets are seen around Nokha, Nagaur, Osian, Sheo and Chohtan which occupy large areas in the central tract parallel to length of the basin and Aravali trend.

Depth to water level (m bgl)	District wise area coverage (sq km)*												Total Area (sq km)
	Ajmer	Barmer	Bikaner	Churu	Ganganagar	Hanumangarh	Jaisalmer	Jalor	Jhunjhunun	Jodhpur	Nagaur	Sikar	
< 10	-	501.9	216.5	4.7	4,086.8	1,508.7	476.7	-	-	184.8	36.0	-	7,016.1
10 - 20	-	3,041.3	3,200.2	1,308.8	4,900.3	5,412.3	4,088.9	52.0	0.1	1,486.1	404.8	27.3	23,922.1
20 - 30	-	3,427.7	6,295.3	3,361.4	1,182.9	2,079.2	4,628.6	23.7	141.2	2,691.0	1,337.7	62.1	25,230.8
30 - 40	48.4	3,139.7	647.6	2,580.3	-	-	821.5	16.2	746.0	2,720.6	2,847.4	950.0	14,517.7
40 - 50	-	3,020.6	5,982.1	3,882.1	514.1	869.5	13,826.5	-	789.9	3,067.3	2,979.8	1,287.1	36,219.0
50 - 60	-	2,231.1	3,384.0	1,547.4	-	39.9	4,386.7	-	1,019.7	2,123.4	1,652.3	1,083.0	17,467.5
60 - 70	-	2,574.8	2,648.6	423.1	-	-	3,335.1	-	781.2	2,404.7	1,258.6	829.6	14,255.7
70 - 80	-	1,924.6	1,971.7	312.1	-	-	2,128.9	-	29.6	2,661.5	1,066.1	41.4	10,135.9
80 - 90	-	1,280.7	1,962.2	366.5	-	-	2,232.4	-	-	1,301.4	1,156.5	-	8,299.7
90 - 100	-	668.2	1,944.7	29.0	-	-	1,454.9	-	-	489.5	357.8	-	4,944.1
100 - 110	-	41.0	1,422.5	1.8	-	-	687.8	-	-	56.5	133.6	-	2,343.2
110 - 120	-	-	557.1	-	-	-	352.0	-	-	5.1	79.0	-	993.2
120 - 130	-	-	46.8	-	-	-	61.4	-	-	-	36.4	-	144.6
> 130	-	-	-	-	-	-	5.8	-	-	-	6.0	-	11.8
Total	48.4	21,851.6	30,279.3	13,817.2	10,684.1	9,909.6	38,487.2	91.9	3,507.7	19,191.9	13,352.0	4,280.5	1,65,501.4

* The area covered in the derived maps is less than the total basin area since the hills have been excluded from interpolation/contouring.



WATER TABLE ELEVATION (PRE MONSOON 2010)

OUTSIDE BASIN

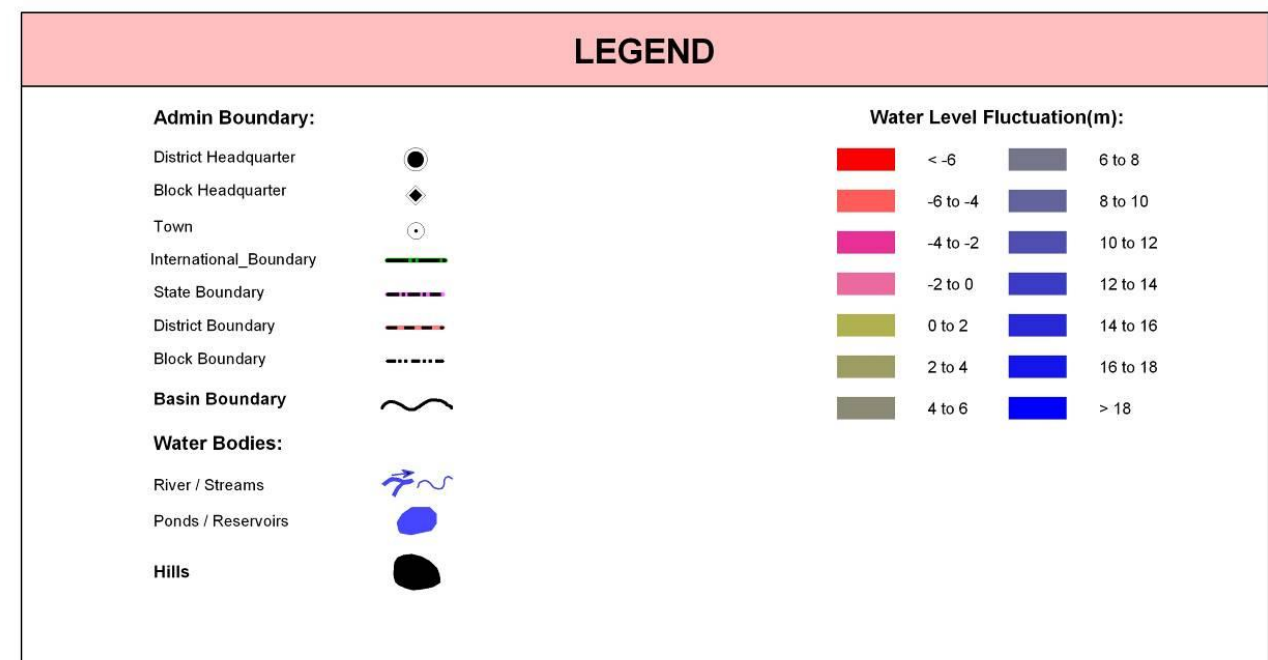
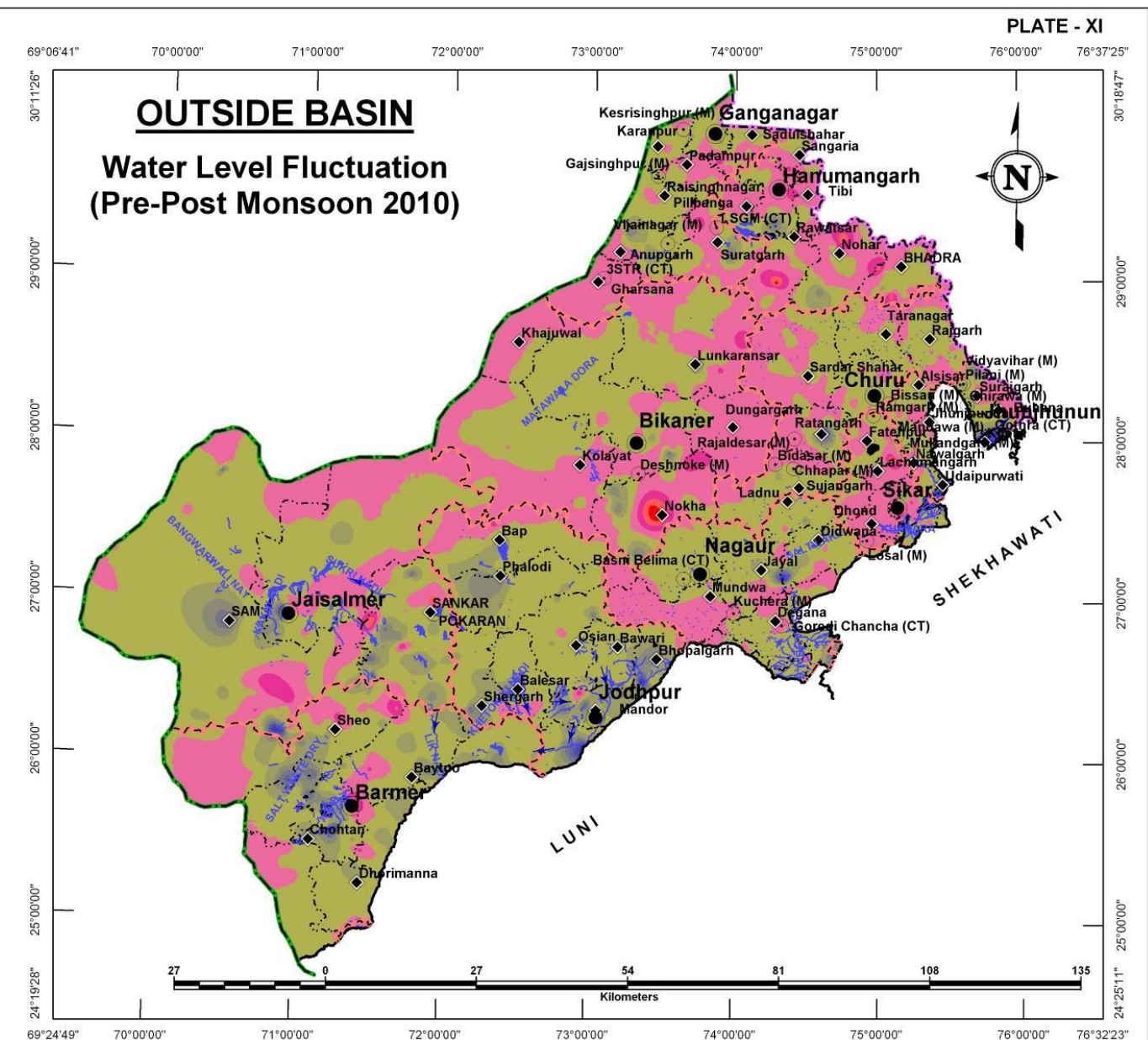
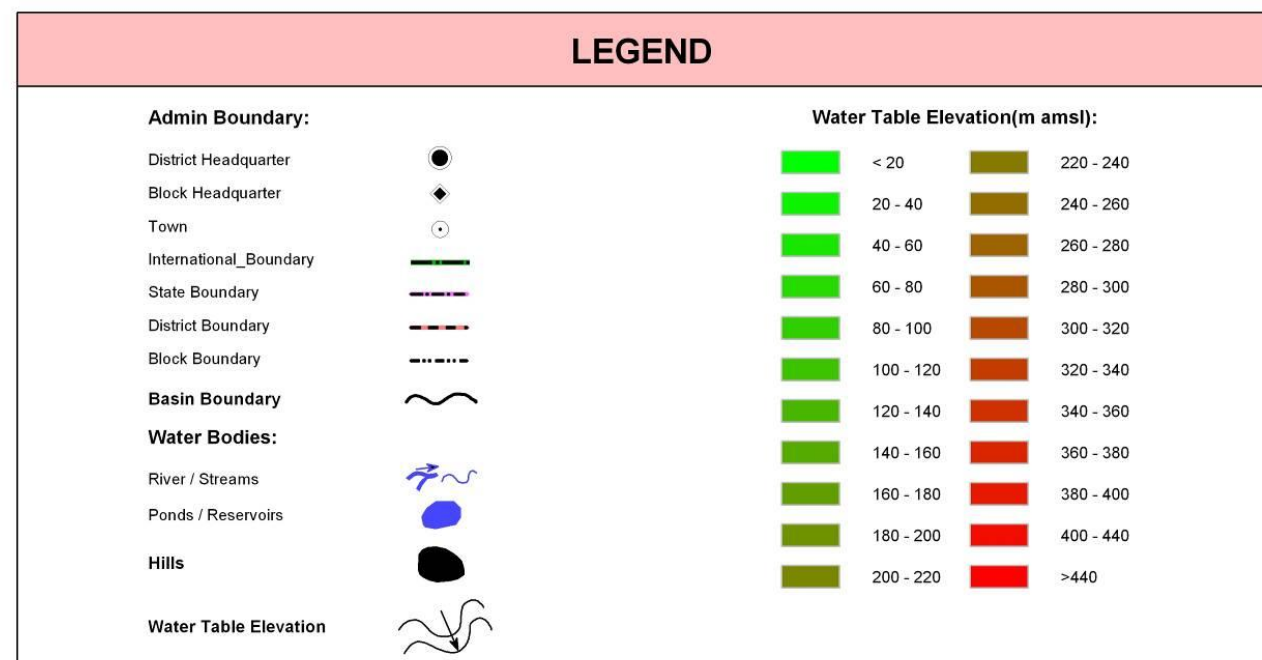
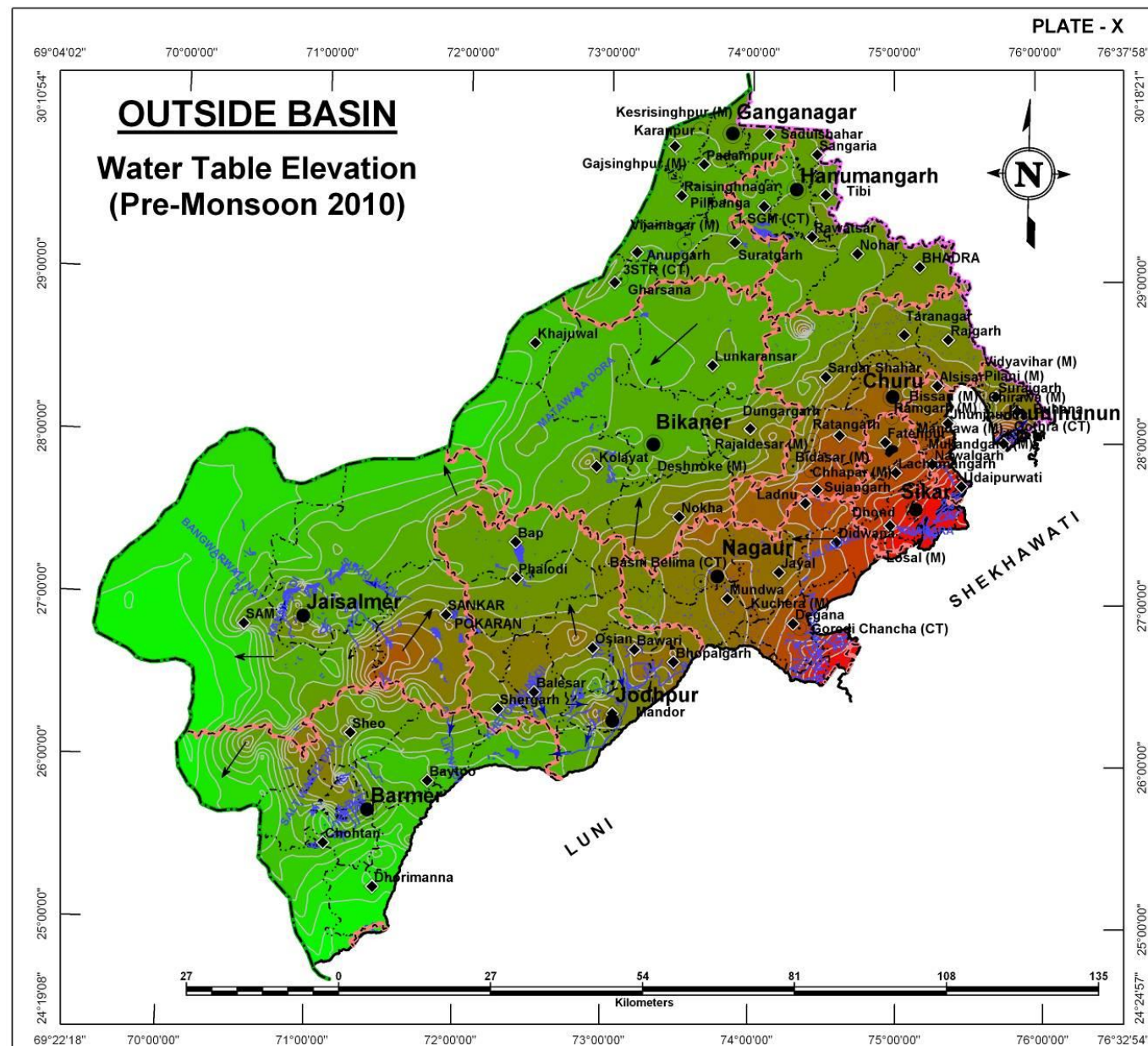
Water table elevation shows large variation ranging from less than 20m above mean sea level in southernmost part of the basin (in Barmer district) to more than 440m amsl (close to Aravali hills in Sikar and Nagaur districts). The ground water flow revealed by the water table contour map is from northeastern part of the basin to west and then towards southwest and southeastern direction draining into Pakistan. There are some local flow reversals as seen to the east of Jaisalmer and northwest of Barmer.

District Name	District wise area coverage (sq km) per Water Table Elevation Range (m amsl)																					Total Area (sq km)	
	< 20	20 - 40	40 - 60	60 - 80	80 - 100	100 - 120	120 - 140	140 - 160	160 - 180	180 - 200	200 - 220	220 - 240	240 - 260	260 - 280	280 - 300	300 - 320	320 - 340	340 - 360	360 - 380	380 - 400	400 - 440		> 440
Ajmer	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	48.4	-	48.4
Barmer	272.3	1,838.5	1,964.2	1,999.2	2,216.4	3,308.5	2,732.8	1,971.9	2,462.4	1,479.5	1,074.6	428.3	78.3	13.8	10.9	-	-	-	-	-	-	-	21,851.6
Bikaner	-	-	-	740.8	1,254.7	2,198.7	6,740.2	9,717.6	3,196.1	2,215.5	1,887.2	1,713.2	430.0	185.3	-	-	-	-	-	-	-	-	30,279.3
Churu	-	-	-	-	-	-	-	645.3	1,268.0	2,056.0	2,599.9	1,854.1	1,548.1	2,089.5	1,114.3	642.0	-	-	-	-	-	-	13,817.2
Ganganagar	-	-	-	-	-	-	1,459.6	6,121.5	3,095.2	7.8	-	-	-	-	-	-	-	-	-	-	-	-	10,684.1
Hanumangarh	-	-	-	-	-	-	-	1,726.8	5,209.6	2,877.5	95.7	-	-	-	-	-	-	-	-	-	-	-	9,909.6
Jaisalmer	52.2	4,232.9	5,251.4	3,859.5	4,243.6	3,215.7	3,576.5	4,964.4	3,000.9	1,393.3	1,161.4	1,888.1	789.6	544.3	313.4	-	-	-	-	-	-	-	38,487.2
Jalor	-	91.9	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	91.9
Jhunjhunun	-	-	-	-	-	-	-	-	-	13.0	328.5	734.7	635.5	495.6	378.0	293.9	157.0	112.4	102.0	99.6	155.3	2.2	3,507.7
Jodhpur	-	-	-	-	-	-	166.9	1,151.0	4,320.0	4,398.0	4,655.6	3,573.7	903.1	23.6	-	-	-	-	-	-	-	-	19,191.9
Nagaur	-	-	-	-	-	-	-	-	1.8	205.9	929.8	1,881.8	3,318.0	1,645.2	1,225.2	1,994.8	1,009.5	310.6	243.5	243.5	337.0	5.4	13,352.0
Sikar	-	-	-	-	-	-	-	-	-	-	-	-	-	539.2	614.9	701.9	467.0	379.7	365.8	302.2	852.7	57.1	4,280.5
Total	324.5	6,163.3	7,215.6	6,599.5	7,714.7	8,722.9	14,676.0	26,298.5	22,554.0	14,646.5	12,732.7	12,073.9	7,702.6	5,536.5	3,656.7	3,632.6	1,633.5	802.7	711.3	645.3	1,393.4	64.7	1,65,501.4

WATER LEVEL FLUCTUATION (PRE TO POST MONSOON 2010)

Ground water level fluctuation also shows very wide range as seen from -6m to > 18m between the water levels measured in pre to post monsoon levels. There are three high exploitation area as seen by high -ve values around Nokha, east and south of Rawatsar and similarly very high water level rise (to the tune of +18m) is also a localized phenomenon seen around Mukandgarh in Jhunjhunun district. Otherwise, in general the water level fluctuation in the basin has been in the -2 to +4 meter range covering almost 95% of the area.

District Name	District wise area coverage (sq km) within fluctuation range (m)														Total Area (sq km)
	< -6	-6 to -4	-4 to -2	-2 to 0	0 to 2	2 to 4	4 to 6	6 to 8	8 to 10	10 to 12	12 to 14	14 to 16	16 to 18	> 18	
Ajmer	-	-	-	-	-	0.6	35.3	12.5	-	-	-	-	-	-	48.4
Barmer	-	31.5	238.7	5,020.5	11,996.3	3,441.7	868.2	198.3	41.8	11.1	3.5	-	-	-	21,851.6
Bikaner	78.4	168.4	617.7	15,585.2	13,188.5	482.9	152.2	6.0	-	-	-	-	-	-	30,279.3
Churu	-	-	36.6	5,737.0	7,801.8	125.0	49.0	29.6	18.4	10.9	6.0	2.4	0.5	-	13,817.2
Ganganagar	-	-	67.7	4,872.9	5,144.8	483.2	96.2	15.5	3.8	-	-	-	-	-	10,684.1
Hanumangarh	4.4	91.0	597.9	5,224.0	3,480.0	404.6	91.7	16.0	-	-	-	-	-	-	9,909.6
Jaisalmer	-	64.3	424.8	9,290.4	24,333.6	3,138.8	867.7	243.4	66.5	39.5	18.2	-	-	-	38,487.2
Jalor	-	-	-	82.8	9.1	-	-	-	-	-	-	-	-	-	91.9
Jhunjhunun	-	-	-	1,085.5	1,914.3	238.4	119.5	68.5	29.6	12.3	10.2	12.1	16.0	1.3	3,507.7
Jodhpur	-	-	24.5	1,747.9	12,451.2	3,517.3	1,118.5	184.4	72.5	46.7	23.4	3.5	1.9	0.1	19,191.9
Nagaur	-	-	1.0	3,984.7	8,720.6	455.9	93.2	50.4	20.4	13.3	7.8	4.3	0.4	-	13,352.0
Sikar	3.8	17.2	86.5	2,605.6	1,510.0	32.3	18.7	6.0	0.4	-	-	-	-	-	4,280.5
Total	86.6	372.4	2,095.4	55,236.5	90,550.2	12,320.7	3,510.2	830.6	253.4	133.8	69.1	22.3	18.8	1.4	1,65,501.4



ELECTRICAL CONDUCTIVITY DISTRIBUTION

OUTSIDE BASIN

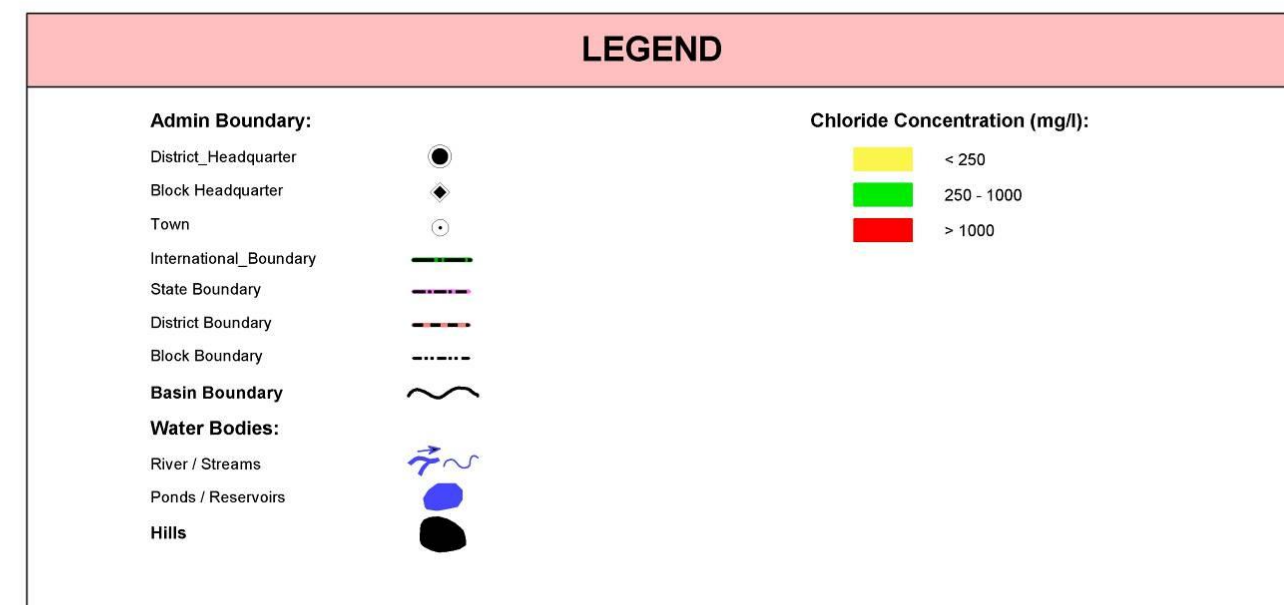
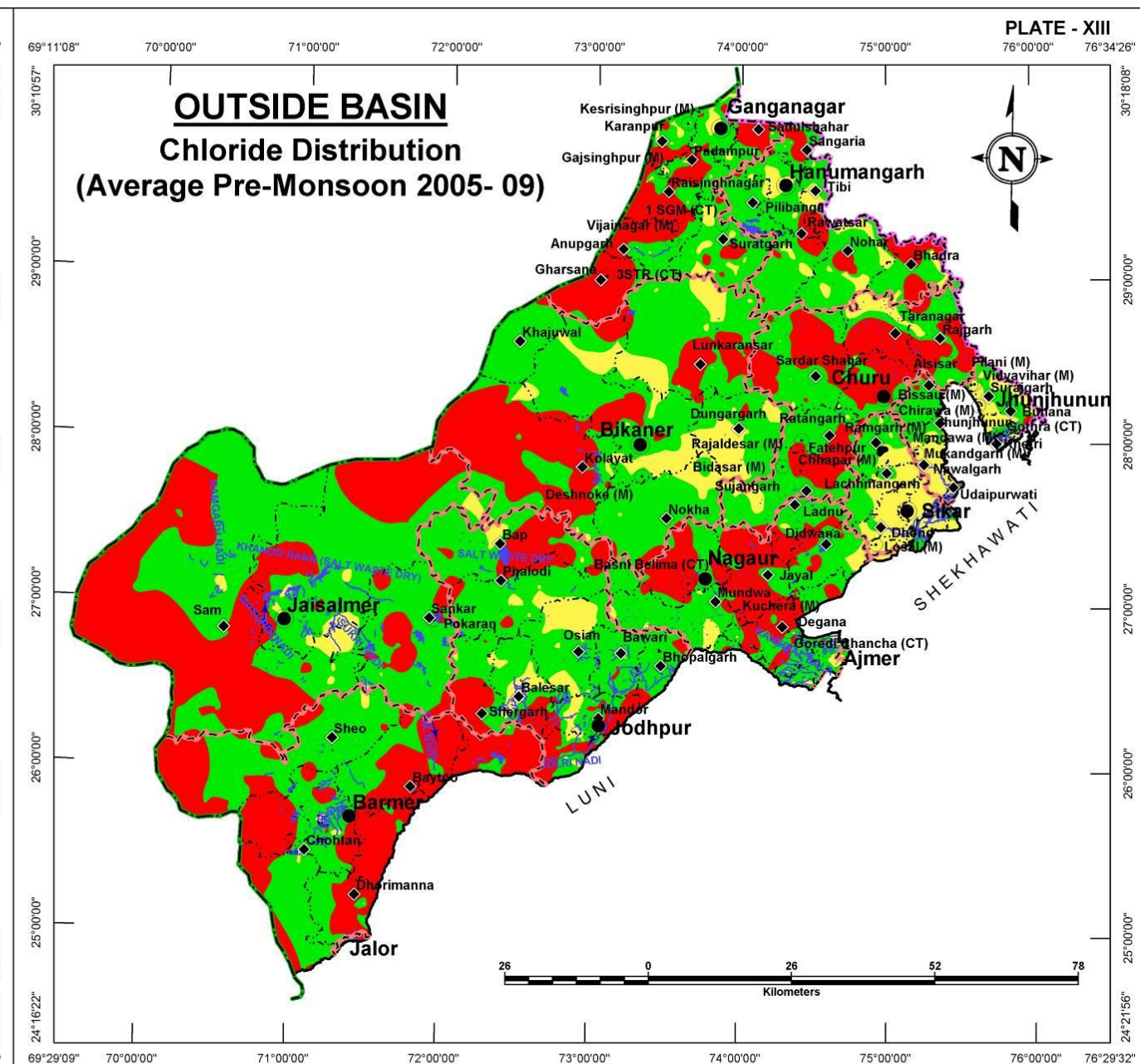
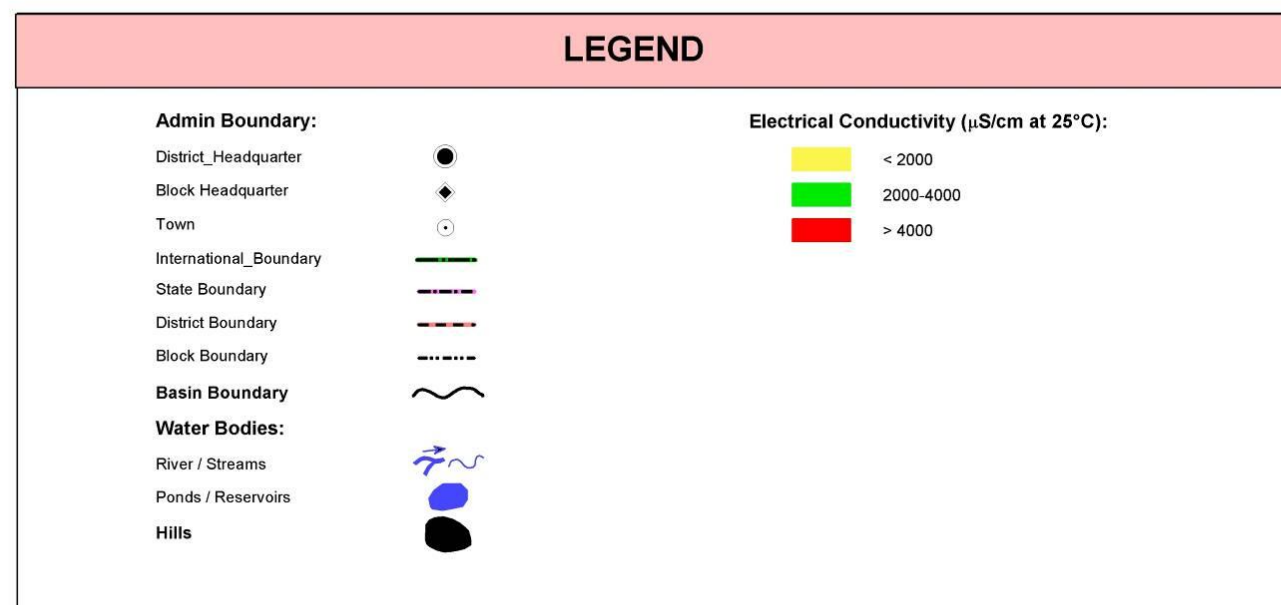
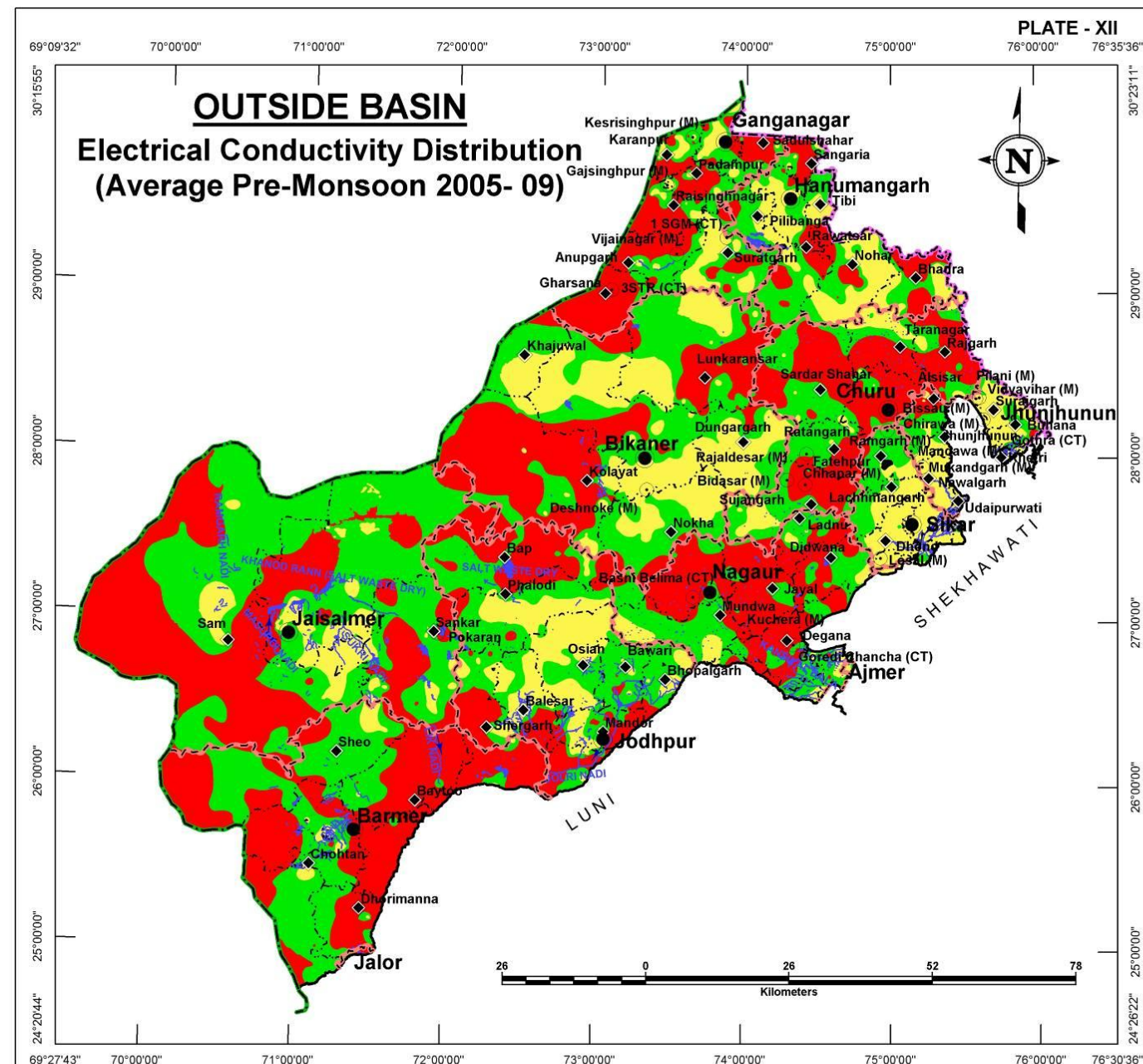
The Electrical Conductivity (at 25°C) distribution map is presented in Plate XII reveals a mixed distribution of high and low EC pockets. Low EC of ground water i.e., below 2000 $\mu\text{S}/\text{cm}$ (represented by Yellow coloured regions) in the map is distributed in patches to the east of Jaisalmer, west and north of Osian, east of Bikaner, around Sikar, and in the vicinity of Hanumangarh. The high EC areas (i.e., $\text{EC} > 4000 \mu\text{S}/\text{cm}$ shown in red colour) are also equally widespread as seen to the east of Barmer and Nagaur; to the north of Churu and west of Jaisalmer, Bikaner and around Anupgarh. The high salinity water is spread over almost 43% of the basin area whereas the low EC areas occupy only 18%.

Electrical Conductivity Ranges ($\mu\text{S}/\text{cm}$ at 25°C) (Ave. of years 2005-09)	District wise area coverage (sq km)																						Total Area (sq km)		
	Ajmer		Barmer		Bikaner		Churu		Ganganagar		Hanumangarh		Jaisalmer		Jalor		Jhunjhunun		Jodhpur		Nagaur			Sikar	
	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age		Area	% age
< 2000	34.1	70.4	452.8	2.1	8,914.0	29.4	1,095.3	7.9	1,445.5	13.5	3,323.9	33.5	2,991.8	7.8	-	-	2,143.5	61.1	4,966.2	25.9	1,016.4	7.6	2,590.0	60.5	28,973.5
2000-4000	14.3	29.6	8,441.6	38.6	11,010.4	36.4	4,947.7	35.8	3,552.3	33.3	4,001.9	40.4	18,205.1	47.3	39.2	42.7	986.8	28.1	8,103.6	42.2	4,890.8	36.6	1,508.0	35.2	65,701.7
> 4000	-	-	12,957.2	59.3	10,354.9	34.2	7,774.2	56.3	5,686.3	53.2	2,583.8	26.1	17,290.3	44.9	52.7	57.3	377.4	10.8	6,122.1	31.9	7,444.8	55.8	182.5	4.3	70,826.2
Total	48.4	100.0	21,851.6	100.0	30,279.3	100.0	13,817.2	100.0	10,684.1	100.0	9,909.6	100.0	38,487.2	100.0	91.9	100.0	3,507.7	100.0	19,191.9	100.0	13,352.0	100.0	4,280.5	100.0	1,65,501.4

CHLORIDE DISTRIBUTION

High chloride concentration in ground water also renders it unsuitable for domestic and other purposes. Green coloured regions i.e., areas representing moderate concentration (250-1000 mg/l) occupy large areas of the basin which is approximately 55% of the basin area. This is followed by high chloride concentration areas (>1000 mg/l, red colour) that cover about 37% of the basin area. Together these areas add up to 92% of the basin rendering these areas to be of limited use for domestic purposes. Thus, only 8% of the basin area located around Sikar, east and north of Bikaner, and Osian and a small patch to the east of Jaisalmer are suitable for all purposes.

Chloride Ranges (mg/l) (Ave. of years 2005-09)	District wise area coverage (sq km)																								Total Area (sq km)
	Ajmer		Barmer		Bikaner		Churu		Ganganagar		Hanumangarh		Jaisalmer		Jalor		Jhunjhunun		Jodhpur		Nagaur		Sikar		
	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	
< 250	13.1	27.0	103.6	0.4	3,962.5	13.1	692.3	5.0	484.8	4.5	1,128.7	11.4	940.5	2.4	-	-	1,530.1	43.6	1,937.1	10.1	594.7	4.5	2,466.4	57.6	13,853.8
250 - 1000	35.3	73.0	10,171.5	46.6	17,488.0	57.7	6,706.2	48.5	5,139.6	48.1	6,790.4	68.5	20,518.0	53.3	44.1	48.0	1,856.3	52.9	12,377.8	64.5	7,326.8	54.8	1,757.1	41.1	90,211.1
> 1000	-	-	11,576.5	53.0	8,828.8	29.2	6,418.7	46.5	5,059.7	47.4	1,990.5	20.1	17,028.7	44.3	47.8	52.0	121.3	3.5	4,877.0	25.4	5,430.5	40.7	57.0	1.3	61,436.5
Total	48.4	100.0	21,851.6	100.0	30,279.3	100.0	13,817.2	100.0	10,684.1	100.0	9,909.6	100.0	38,487.2	100.0	91.9	100.0	3,507.7	100.0	19,191.9	100.0	13,352.0	100.0	4,280.5	100.0	1,65,501.4



FLUORIDE DISTRIBUTION

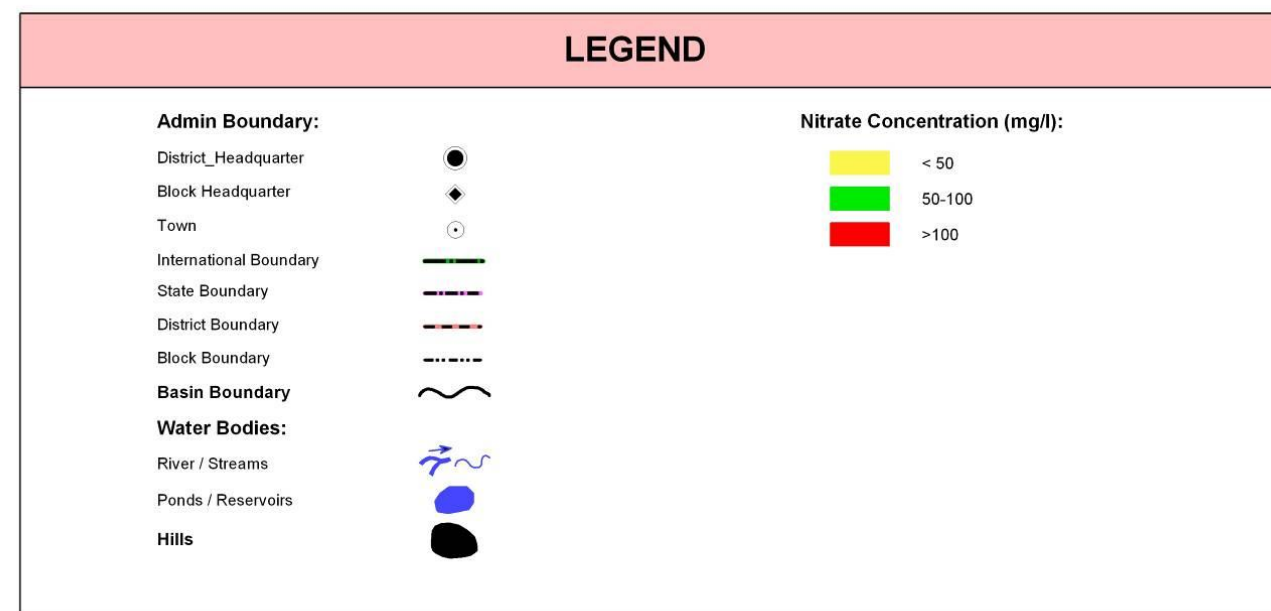
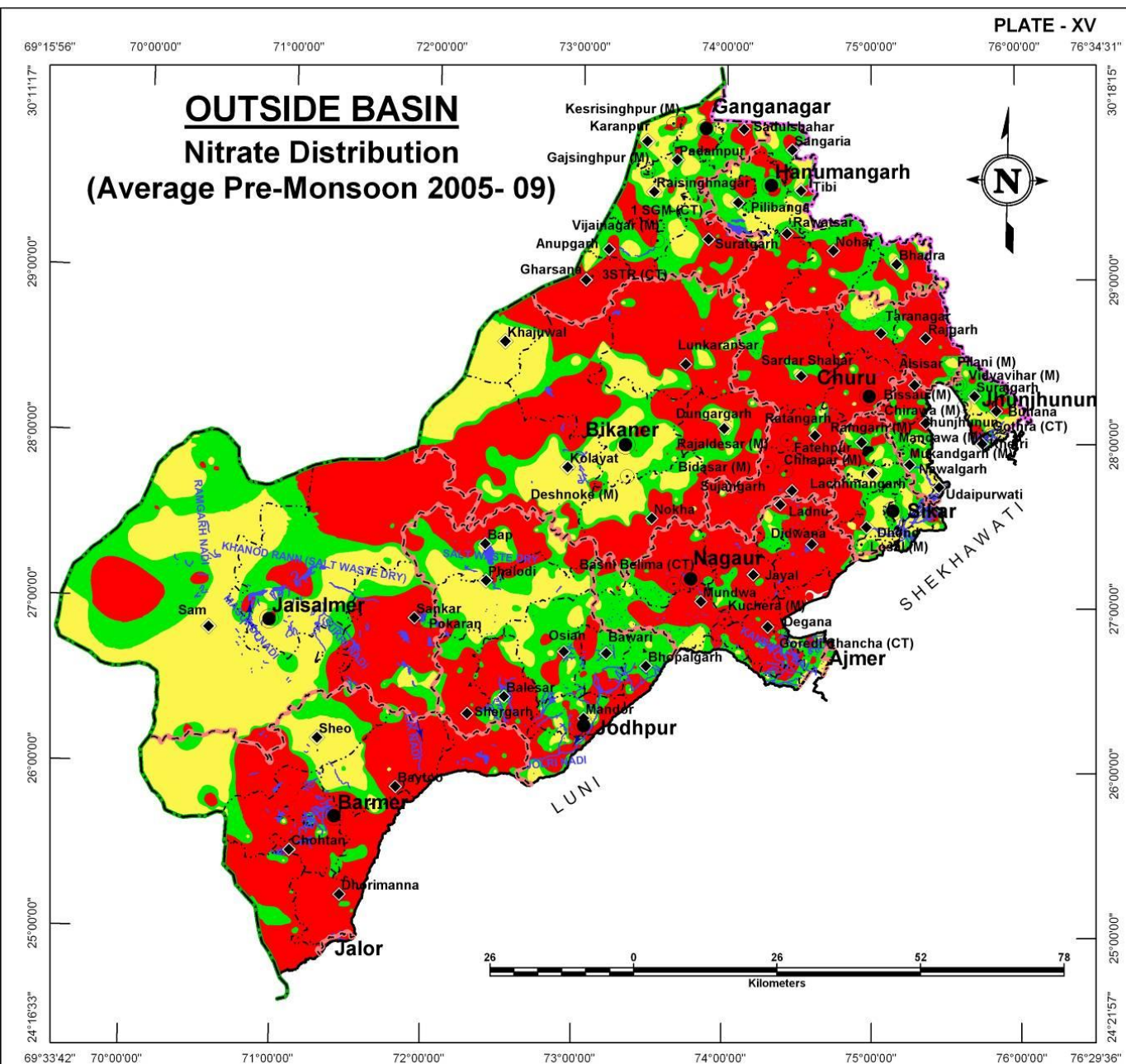
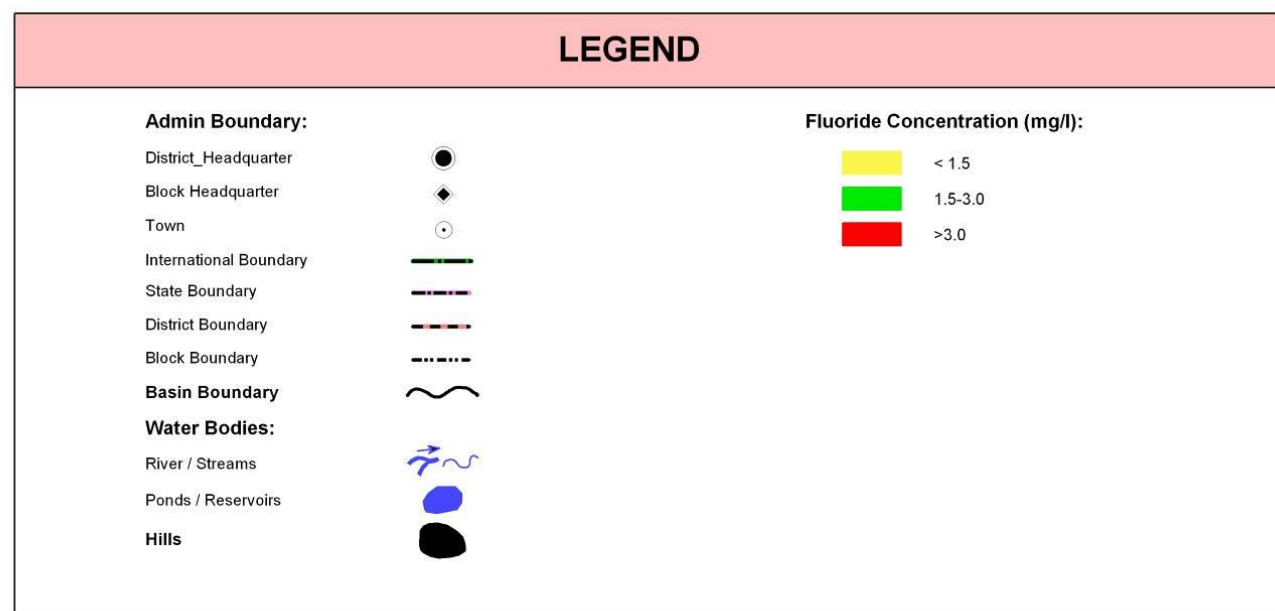
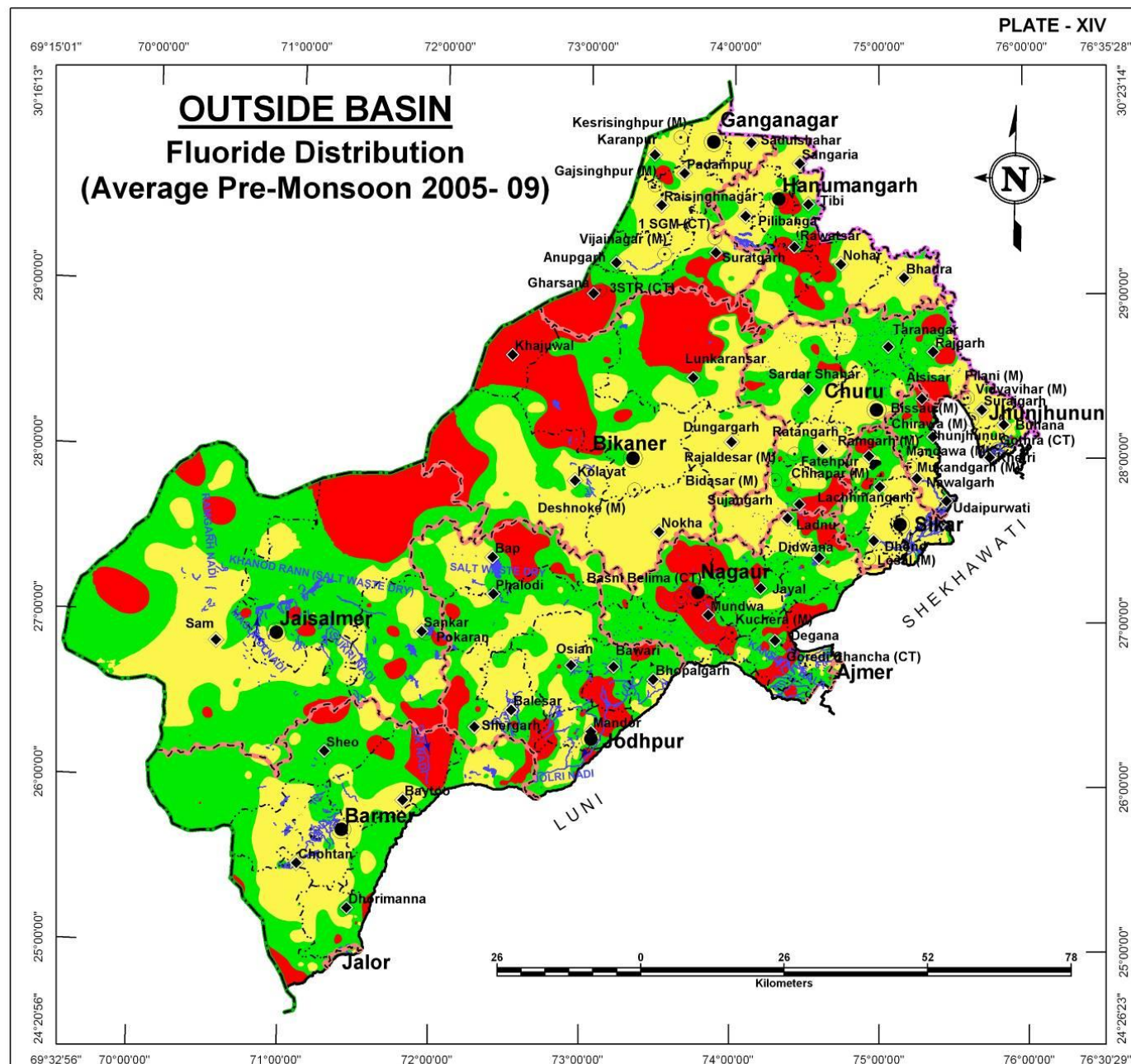
The Fluoride concentration map (Plate – XIV) reveals the border areas with Pakistan, in the northwestern part of the basin to have high fluoride concentration (>3 mg/l) in ground water coupled with some areas in the eastern part of the basin in the Jodhpur-Nagaur-Churu belt in addition to some very small pockets scattered over the basin. Low fluoride (<1.5 mg/l) areas are widespread in the central part of the basin to the east of Bikaner and as large patches around Ganganagar, Hanumangarh, Churu-Bikaner and to the south and west of Osian, Jaisalmer and Barmer. Overall, less than 40% area of the basin falls within low fluoride zone that is suitable for domestic purposes.

Fluoride Ranges (mg/l) (Ave. of years 2005-09)	District wise area coverage (sq km)																						Total Area (sq km)		
	Ajmer		Barmer		Bikaner		Churu		Ganganagar		Hanumangarh		Jaisalmer		Jalor		Jhunjhunun		Jodhpur		Nagaur			Sikar	
	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age		Area	% age
< 1.5	30.7	63.3	7,236.5	33.1	14,853.7	49.1	6,437.6	46.6	6,134.2	57.4	5,553.2	56.0	10,816.1	28.1	14.6	16.0	2,243.6	64.0	6,677.7	34.8	2,122.8	15.9	2,506.8	58.6	64,627.5
1.5-3.0	17.7	36.7	12,261.4	56.1	6,780.3	22.3	5,992.4	43.4	2,415.7	22.6	2,818.9	28.5	20,388.3	53.0	77.3	84.0	962.9	27.4	9,230.2	48.1	7,421.3	55.6	1,442.2	33.7	69,808.6
> 3.0	-	-	2,353.7	10.8	8,645.3	28.6	1,387.2	10.0	2,134.2	20.0	1,537.5	15.5	7,282.8	18.9	-	-	301.2	8.6	3,284.0	17.1	3,807.9	28.5	331.5	7.7	31,065.3
Total	48.4	100.0	21,851.6	100.0	30,279.3	100.0	13,817.2	100.0	10,684.1	100.0	9,909.6	100.0	38,487.2	100.0	91.9	100.0	3,507.7	100.0	19,191.9	100.0	13,352.0	100.0	4,280.5	100.0	165,501.4

NITRATE DISTRIBUTION

High nitrate concentration in ground water renders it unsuitable for agriculture purposes. Plate XV shows distribution of Nitrate in ground water. A perusal of the map reveals major part i.e., almost 3/4th of the basin has high (>50 mg/l) nitrate concentration in ground water which leaves these areas unsuitable for agriculture purposes. Of the remaining 25% where Nitrate is low, most of the area falls within Thar desert, to the south and west of Jaisalmer which too is unsuitable for agriculture. That leaves only small areas between Bikaner and Pokharan with low Nitrate concentration in ground water which too is by and large treated as dry area akin to desert. So, high nitrate in Outside basin limits the area's suitability for agriculture purposes except a small area in the Ganganagar- Hanumangarh region.

Nitrate Ranges (mg/l) (Ave. of years 2005-09)	District wise area coverage (sq km)																								Total Area (sq km)
	Ajmer		Barmer		Bikaner		Churu		Ganganagar		Hanumangarh		Jaisalmer		Jalor		Jhunjhunun		Jodhpur		Nagaur		Sikar		
	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	Area	% age	
< 50	-	-	4,383.4	20.1	8,499.4	28.1	413.5	3.0	2,114.3	19.8	1,419.6	14.3	15,824.6	41.1	-	-	808.3	23.0	3,764.7	19.6	524.7	3.9	1,180.7	27.6	38,933.2
50-100	41.4	85.6	2,807.4	12.8	5,330.2	17.6	1,462.6	10.6	3,590.2	33.6	3,502.2	35.4	10,417.7	27.1	-	-	1,570.5	44.8	7,318.2	38.1	3,097.2	23.2	1,663.6	38.8	40,801.2
> 100	7.0	14.4	14,660.8	67.1	16,449.7	54.3	11,941.1	86.4	4,979.6	46.6	4,987.8	50.3	12,244.9	31.8	91.9	100.0	1,128.9	32.2	8,109.0	42.3	9,730.1	72.9	1,436.2	33.6	85,767.0
Total	48.4	100.0	21,851.6	100.0	30,279.3	100.0	13,817.2	100.0	10,684.1	100.0	9,909.6	100.0	38,487.2	100.0	91.9	100.0	3,507.7	100.0	19,191.9	100.0	13,352.0	100.0	4,280.5	100.0	165,501.4



DEPTH TO BEDROCK

Plate XVI represents depth to bed rock in meters below ground level. The entire area of the Outside basin is underlain by the hardrock at different depths. The depth to bedrock defines the sub surface topography of the occurrence of massive hardrock beneath alluvial deposits and weathered/fractured hardrock. The bedrock is encountered at shallower depths in the eastern strip of the basin ranging in depth from less than 20m bgl to about 120m bgl. However, in the western and southwestern parts of the basin it attains greater depths reaching upto even 300m bgl especially around Bikaner and southwest of Jaisalmer.

Depth to Bedrock (m bgl)	District wise area coverage (sq km)												Total Area (sq km)
	Ajmer	Barmer	Bikaner	Churu	Ganganagar	Hanumangarh	Jaisalmer	Jalor	Jhunjhun	Jodhpur	Nagaur	Sikar	
< 20	-	-	-	0.6	-	-	-	-	-	10.4	26.4	-	37.4
20-40	-	330.3	-	25.8	-	-	19.8	-	5.0	146.6	354.2	-	881.7
40-60	48.4	1,681.2	12.9	284.9	-	-	44.2	-	50.9	1,262.7	677.9	54.7	4,117.8
60-80	-	1,538.3	18.8	1,055.0	-	-	3,075.2	-	426.1	5,969.8	2,778.7	483.6	15,345.5
80-100	-	2,483.0	35.2	3,446.9	-	12.2	1,699.7	-	1,742.2	4,694.1	4,181.3	1,626.3	19,920.9
100-120	-	4,545.9	339.2	4,988.1	-	305.7	3,832.9	3.5	1,260.5	1,507.8	2,534.5	1,876.3	21,194.4
120-140	-	4,255.7	1,253.0	2,121.8	2,362.3	8,996.1	5,053.4	88.4	23.0	1,375.8	1,639.4	239.6	27,408.5
140-160	-	1,931.8	1,945.1	1,070.6	3,714.8	553.7	6,804.3	-	-	1,422.2	527.7	-	17,970.2
160-180	-	1,220.0	3,786.1	579.4	3,208.4	41.9	6,520.1	-	-	1,086.9	299.2	-	16,742.0
180-200	-	1,815.6	6,167.9	184.1	1,131.5	-	5,448.2	-	-	928.3	173.7	-	15,849.3
200-220	-	1,367.4	13,003.3	40.2	267.1	-	3,805.8	-	-	566.7	92.2	-	19,142.7
220-240	-	528.5	2,986.1	13.4	-	-	1,162.1	-	-	220.6	35.4	-	4,946.1
240-260	-	95.8	656.1	6.4	-	-	530.3	-	-	-	23.3	-	1,311.9
260-280	-	42.5	75.6	-	-	-	284.3	-	-	-	8.1	-	410.5
280-300	-	15.6	-	-	-	-	174.4	-	-	-	-	-	190.0
> 300	-	-	-	-	-	-	32.5	-	-	-	-	-	32.5
Total	48.4	21,851.6	30,279.3	13,817.2	10,684.1	9,909.6	38,487.2	91.9	3,507.7	19,191.9	13,352.0	4,280.5	165,501.4

UNCONFINED AQUIFER

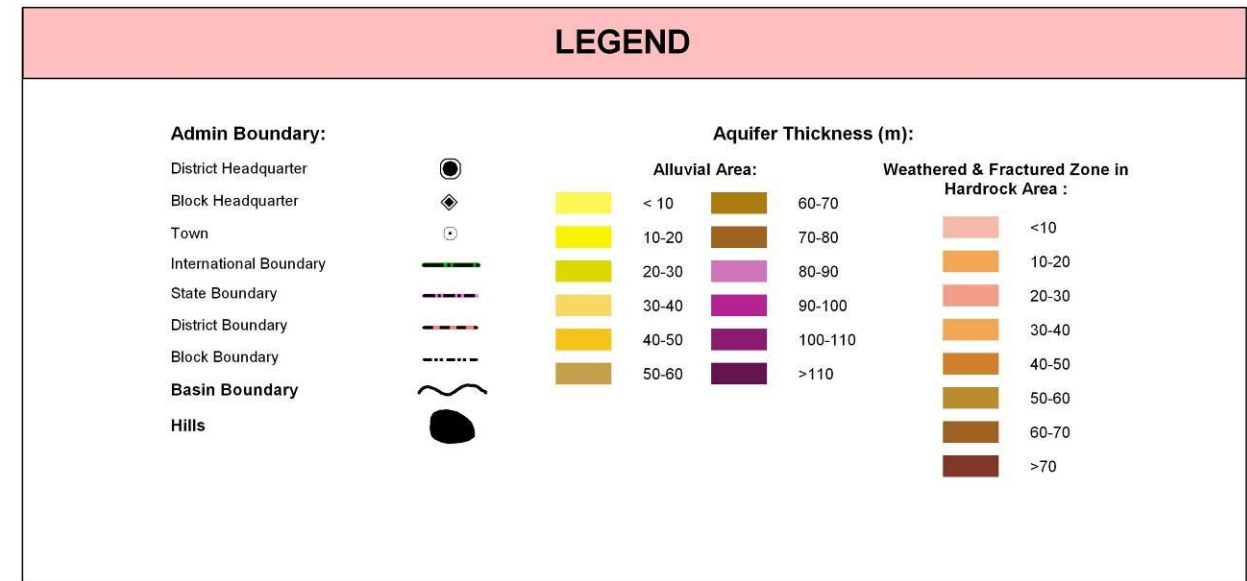
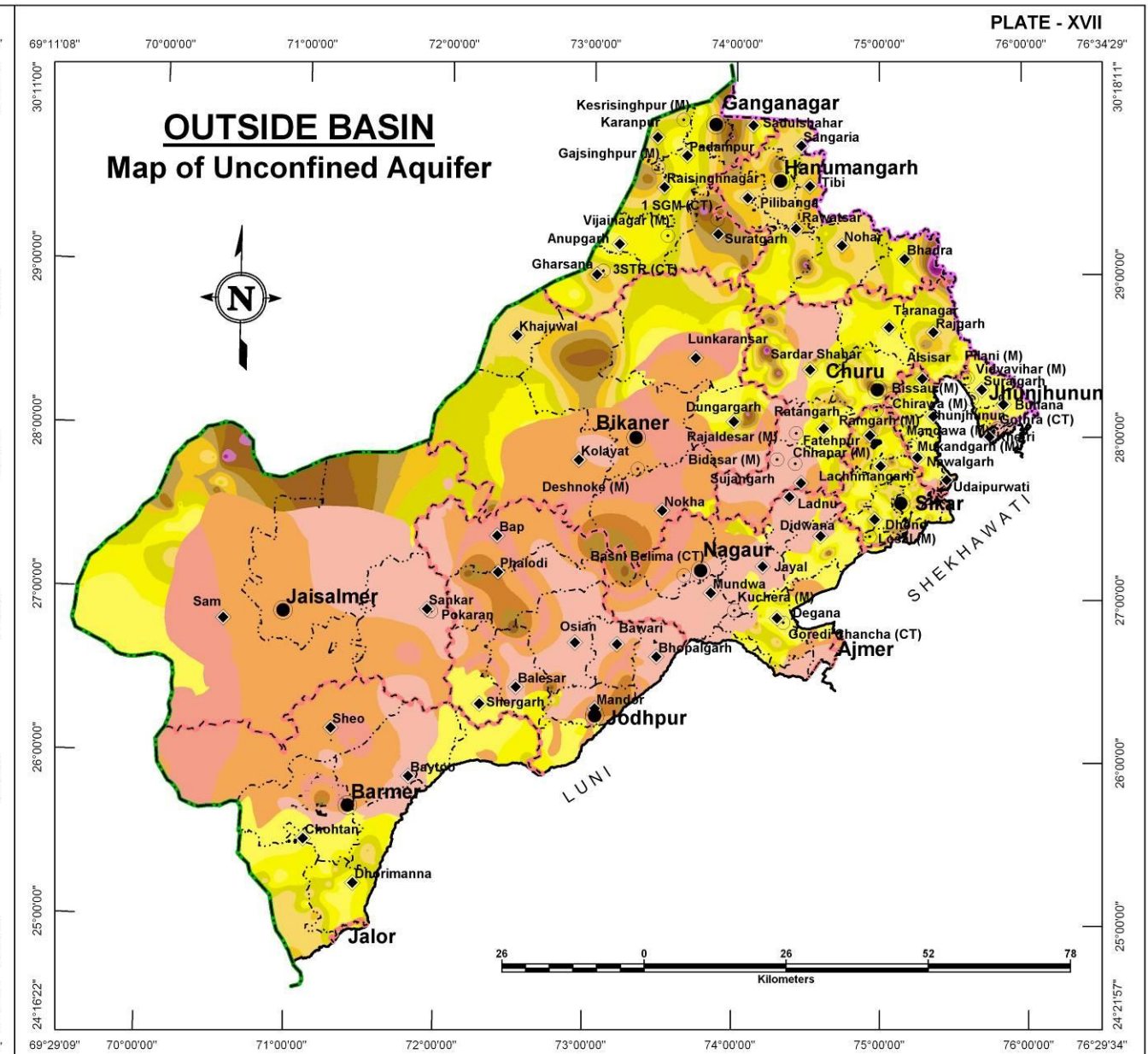
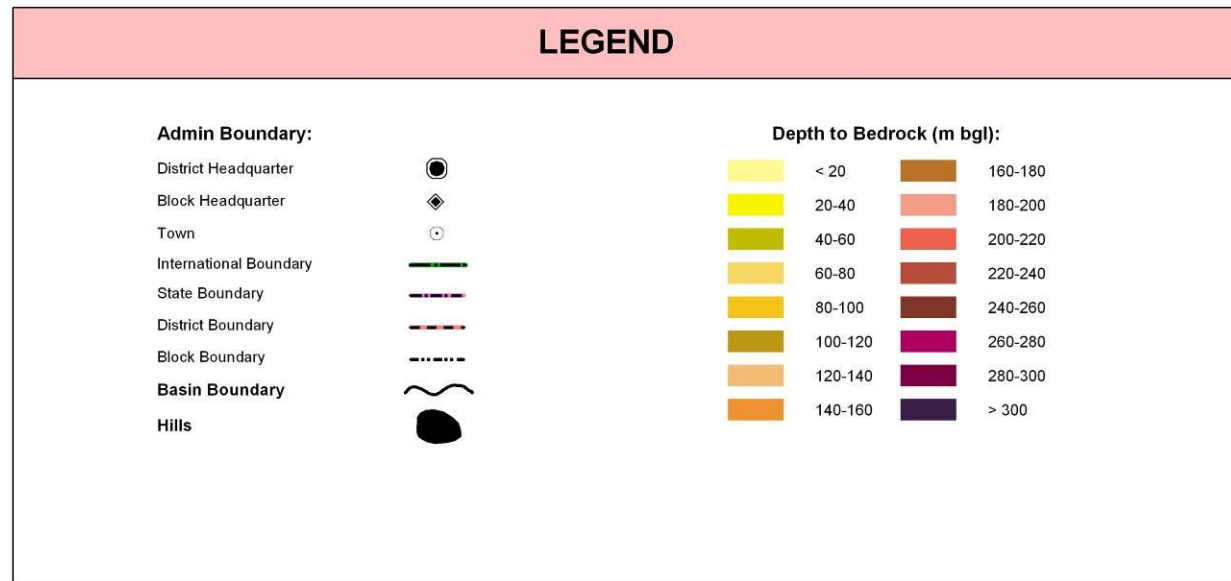
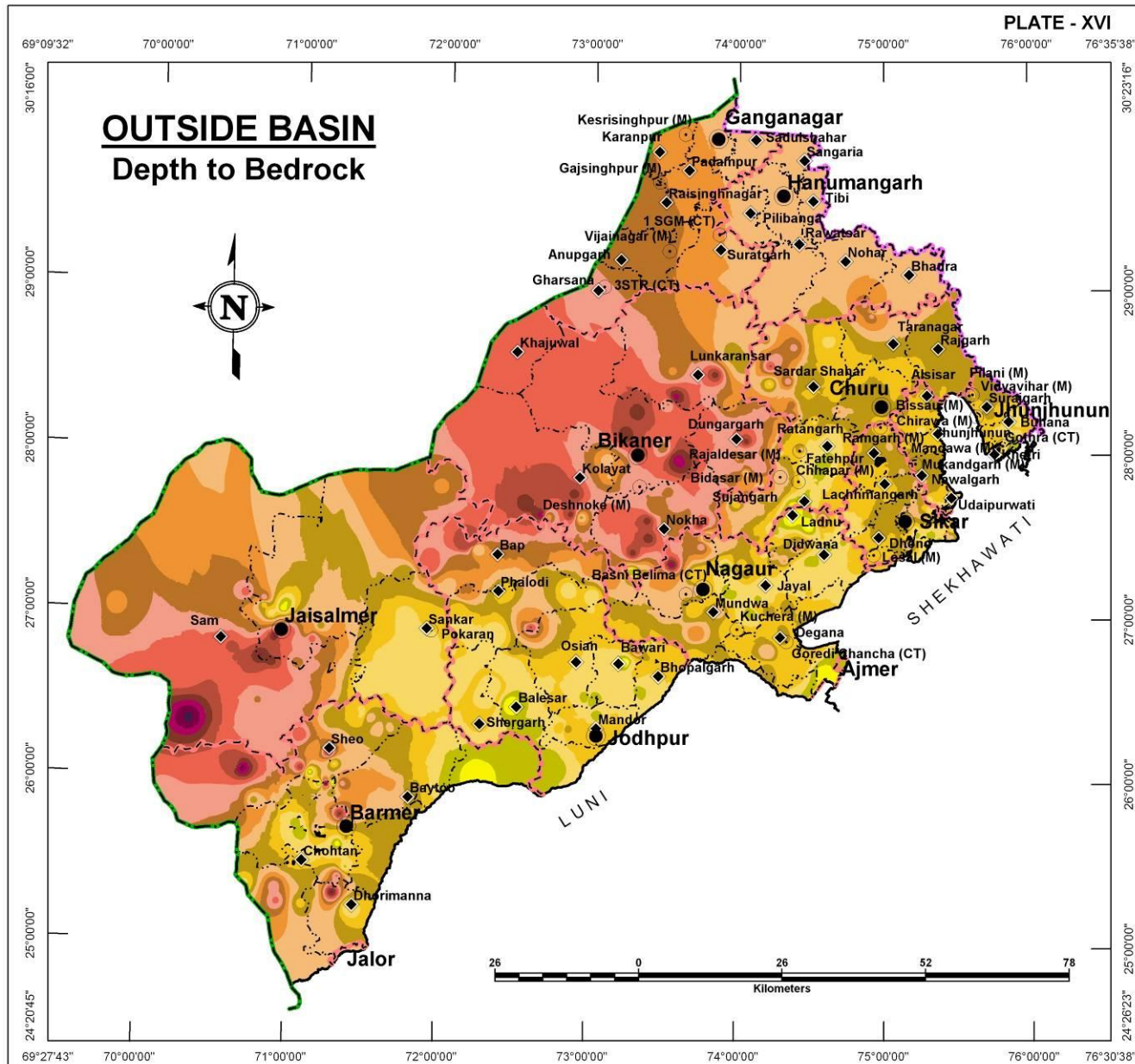
Hydrogeological properties are different for alluvial and hard rock aquifers and therefore, this aquifer has been mapped as two separate regions viz, unconfined aquifers in alluvial and in hard rock areas. Aquifers formed in weathered/fractured zone of hard rocks constitute approximately half of the aquifer area in Outside basin and an equal area has aquifers formed in alluvium. The thickness of alluvial aquifers seem to be much higher than the hardrock aquifers, since the former attains a thickness of upto 120m whereas hardrock aquifers reach a maximum thickness of 70m.

Alluvial areas

Unconfined aquifer Thickness (m)	District wise area coverage (sq km)												Total Area (sq km)
	Ajmer	Barmer	Bikaner	Churu	Ganganagar	Hanumangarh	Jaisalmer	Jalor	Jhunjhun	Jodhpur	Nagaur	Sikar	
< 10	-	4,286.6	1,714.6	2,232.3	442.3	402.3	3,398.4	30.9	1,965.6	1,670.3	1,966.6	765.9	18,875.8
10-20	-	2,405.5	2,836.3	2,911.0	2,634.4	1,422.1	1,679.2	38.8	580.2	546.1	1,059.2	1,103.0	17,215.8
20-30	-	1,291.7	5,041.9	1,881.5	2,224.1	1,801.2	2,460.5	22.2	336.4	89.5	494.7	924.6	16,568.3
30-40	-	1,177.3	2,765.9	1,200.0	1,946.7	3,636.7	216.1	-	211.2	13.2	142.3	645.7	11,955.1
40-50	-	92.2	1,268.5	631.8	1,319.2	1,717.0	1,661.5	-	53.9	-	42.2	417.6	7,203.9
50-60	-	-	887.9	358.4	1,200.3	453.3	1,366.2	-	1.6	-	15.3	158.8	4,441.8
60-70	-	-	600.9	141.8	713.4	200.9	1,375.8	-	-	-	7.2	18.5	3,058.5
70-80	-	-	281.5	59.4	173.5	117.3	1,883.1	-	-	-	4.3	9.1	2,528.2
80-90	-	-	6.0	24.4	30.2	77.2	113.9	-	-	-	1.6	-	253.3
90-100	-	-	-	6.4	-	42.8	7.3	-	-	-	-	-	56.5
100-110	-	-	-	-	-	35.0	-	-	-	-	-	-	35.0
> 110	-	-	-	-	-	3.8	-	-	-	-	-	-	3.8
Total	-	9,253.3	15,403.5	9,447.0	10,684.1	9,909.6	14,162.0	91.9	3,148.9	2,319.1	3,733.4	4,043.2	82,196.0

Hardrock areas:

Unconfined aquifer Thickness (m)	District wise area coverage (sq km)												Total Area (sq km)
	Ajmer	Barmer	Bikaner	Churu	Ganga- nagar	Hanuman- garh	Jaisalmer	Jalor	Jhunjhun	Jodhpur	Nagaur	Sikar	
<10	16.0	2,756.6	1,059.3	3,512.7	-	-	6,965.5	-	236.5	6,748.9	5,832.9	217.9	27,346.3
10-20	32.4	6,470.4	5,658.5	758.3	-	-	12,695.8	-	84.9	4,040.5	2,043.6	16.8	31,801.2
20-30	-	3,039.6	4,470.3	99.2	-	-	4,663.9	-	17.3	2,376.2	812.0	2.6	15,481.1
30-40	-	241.8	2,770.9	-	-	-	-	-	14.0	1,601.1	394.4	-	5,022.2
40-50	-	62.9	612.2	-	-	-	-	-	6.1	1,402.0	240.9	-	2,324.1
50-60	-	23.3	304.6	-	-	-	-	-	-	618.6	263.5	-	1,210.0
60-70	-	3.7	-	-	-	-	-	-	-	83.6	31.3	-	118.6
> 70	-	-	-	-	-	-	-	-	-	1.9	-	-	1.9
Total	48.4	12,598.3	14,875.8	4,370.2	-	-	24,325.2	-	358.8	16,872.8	9,618.6	237.3	83,305.4

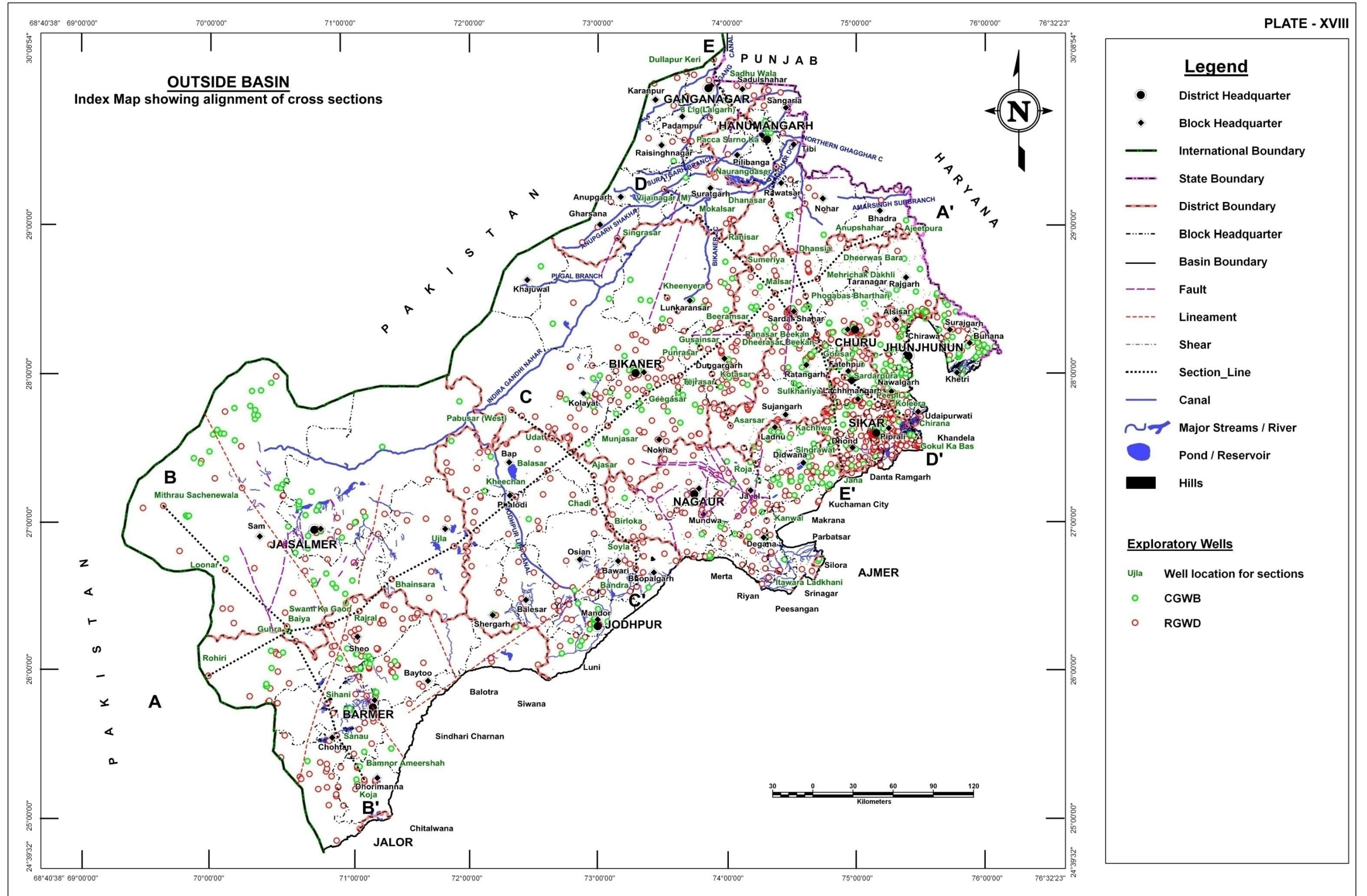


CROSS SECTIONS

OUTSIDE BASIN

Several hydrogeologic cross sections have been drawn to better depict the sub-surface distribution of lithology. These sections have been overlaid with geological maps and structural faults if there are any have been transferred for verification of their impact on sub-surface material disposition. The alignment of the cross sections is shown in Plate – XVIII and corresponding sections are presented in Plates – XIX to XXIII. The broad alignment of the sections is as given below:

Name of Section Line	Orientation
Section AA'	SW-NE
Section BB'	NW-SE
Section CC'	NW-SE
Section DD'	NW-SE
Section EE'	N-S



CROSS SECTIONS

OUTSIDE BASIN

Section A-A':

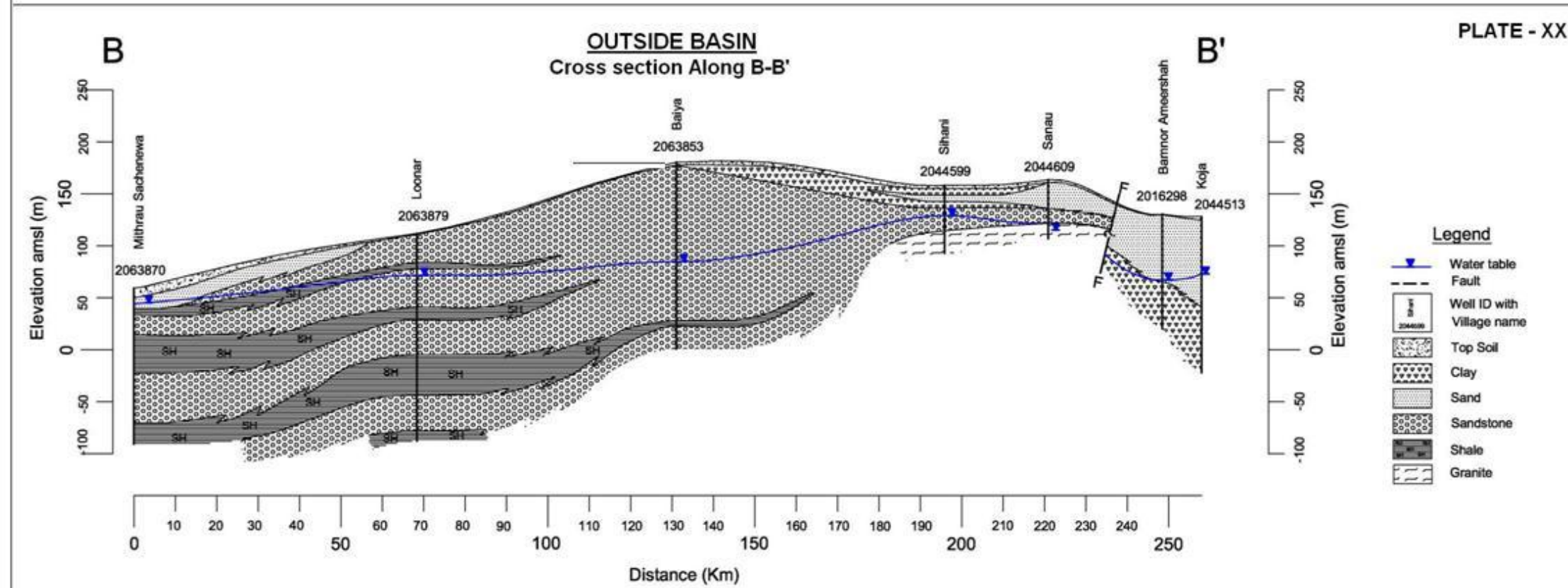
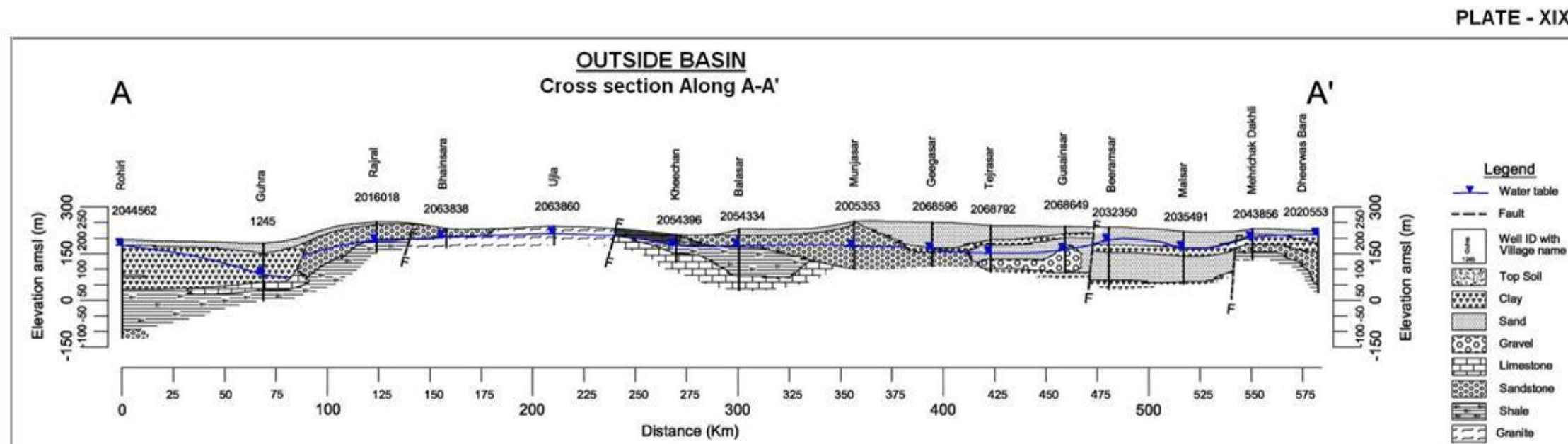
The A-A' section is the longest of the sections plotted in the area covering a length of 585 km and trends in SW-NE direction, cutting across the basin (Plate – XIX). Lithologs of 18 boreholes and their surrounding information is taken into consideration while preparing this cross section. This section presents layers of sand, gravel and clay along with sandstone, limestone, shale, weathered and fractured zones of Granite. Four tentative faults are indicated in the section which limits the extent of granites in the western pair of faults and the second pair in the east demarcates a depression in sandy material bounded by Sandstones on either side. The cross section further indicates that in the northeastern part intercalated layer of sand and clay underlain by sandstone and shale whereas, granite, shale and limestone constitute the bedrock in the southern part.

The water level varies from 95 m amsl to 210 m amsl following the surface topography as observed from the 2010 pre monsoon season.

Section B-B':

The B-B' section is 250 km long (Plate – XX) trending NW-SE. Lithologs of 7 boreholes and their surrounding information is taken into consideration while preparing this cross section. The section depicts sand and clay layers of alluvium aquifers and sandstone and shale as hard rock aquifer. Granite is encountered in two wells in the southeastern part of the section whose extent is apparently delimited by a tentative fault to the east of which a significant increase in thickness of alluvial cover (sand and clay) is noticed. In the northwestern part however, intercalated layers of shale and sandstone are noticed along with some alluvial cover and thin top soil.

It is observed from pre monsoon 2010 data that the ground water level varies from 50m amsl to 130m amsl.



CROSS SECTIONS

OUTSIDE BASIN

Section C-C':

The C-C' section is 180 km long trending NW-SE in the central part of the basin (Plate – XXI). Lithologs of 7 boreholes and their surrounding information is taken into consideration while preparing this cross section. This cross section shows the presence of thick sandstone layer in the northwest terminated against a tentative fault. The central part appears to be a fault bound depression with thick alluvial material and sandstone- shale interlayering. Further southeast, again thick sandstone layer appears with thin shale bed intercalation and a lowermost limestone layer. A tentatively fault bound schist layer also appears in the middle of the section which terminates against the sandstone/shale/limestone rocks on either side.

The water level varies from 150m amsl to 300m amsl as per the data of pre monsoon 2010.

Section D-D':

The D-D' section is about 280 km long trending in NW-SE direction (Plate – XXII) selected in the northeastern part of the basin. Lithologs of 12 boreholes and their surrounding information is taken into consideration while preparing this cross section. It depicts a predominantly alluvium dominant lithology in the sub-surface. The alluvium is alternating layers of clay and sand resting on a hardrock basement formed by phyllite, quartzite and schist. There appears to be a faulted contact between the older rocks in the southeast (phyllite, schist and quartzite) and younger rocks in the northwest made up of limestone, shale and sandstone.

It is observed from pre monsoon 2010 data that the ground water table varies from 150 m amsl to 440 m amsl.

PLATE - XXI

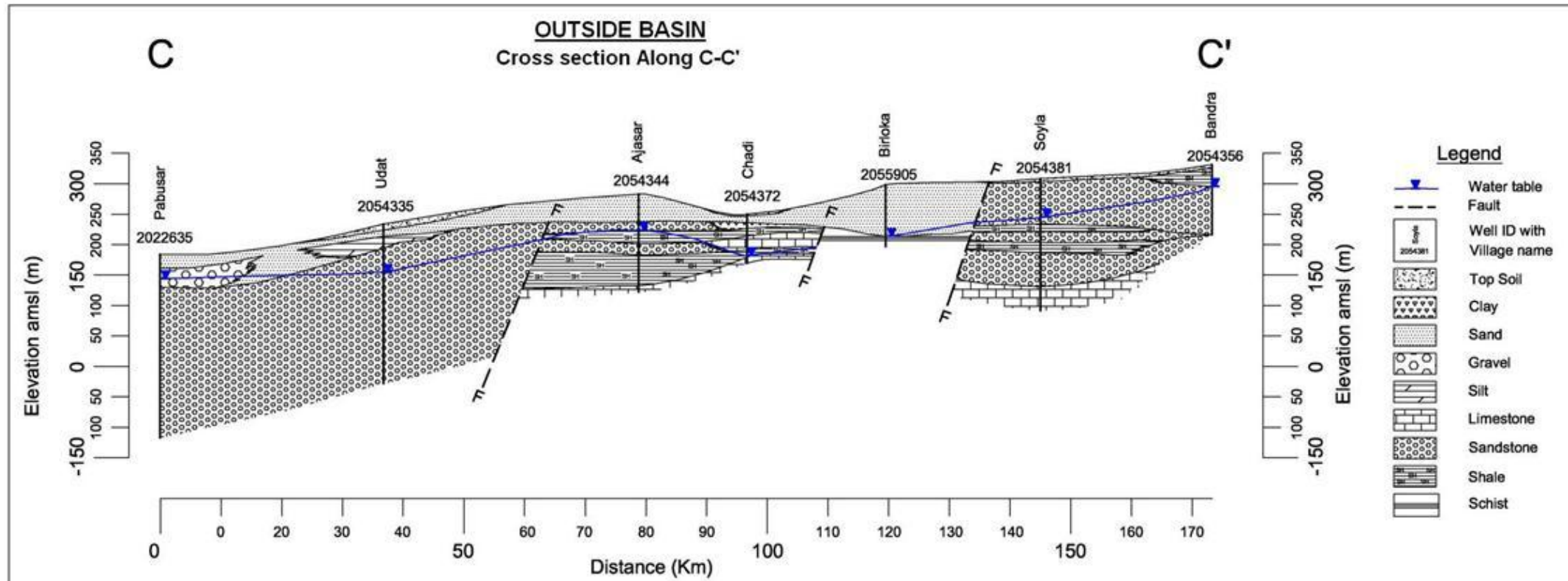
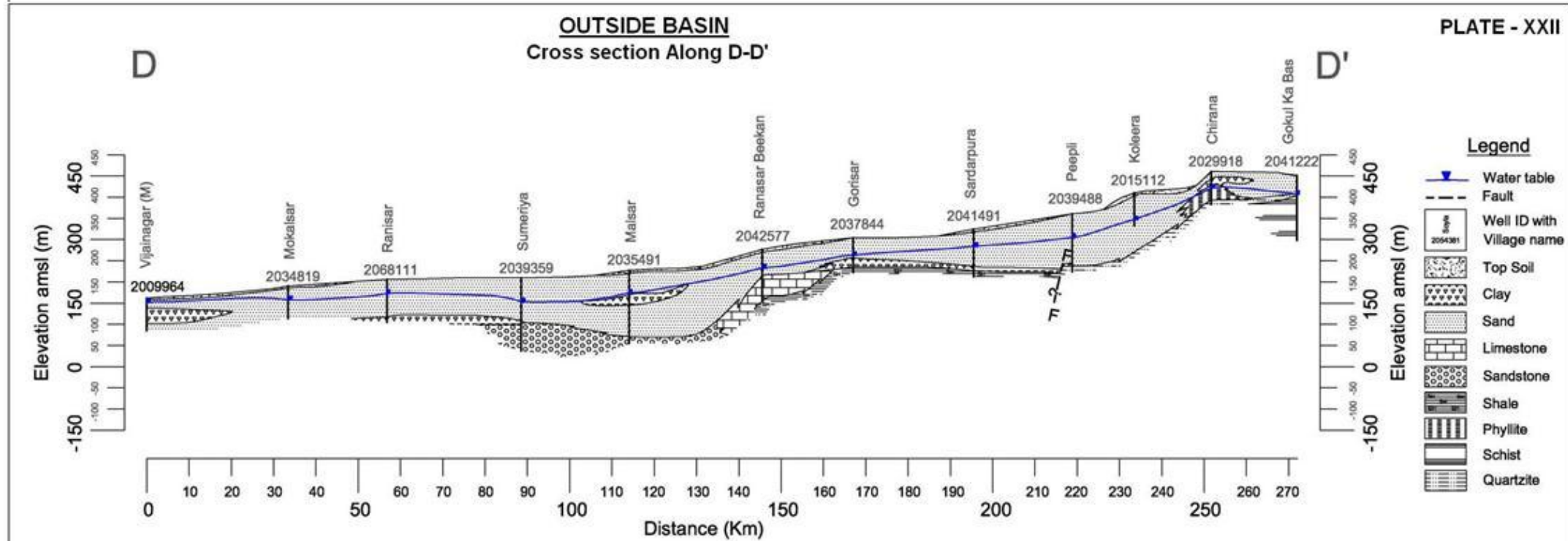


PLATE - XXII



CROSS SECTIONS

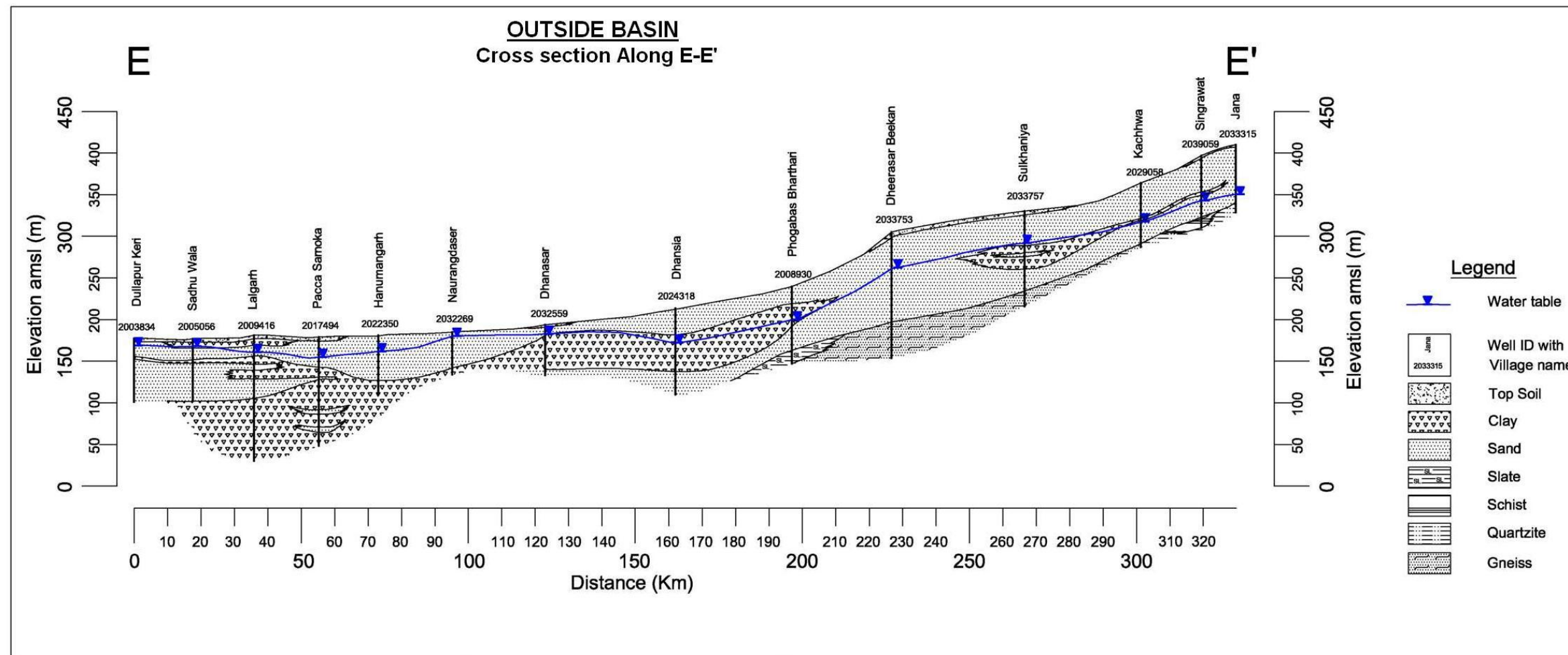
OUTSIDE BASIN

Section E-E’:

The E-E’ section is 350 km long trending in NW-SE direction (Plate – XXIII) selected in the north eastern part of the basin which takes lithologs of 14 boreholes into consideration and their surrounding information while preparing this cross section. The topography steeply rises towards east while the western part is a level plain. Multiple layers of clay and sand predominates in the sub-surface to varying thicknesses from about 50m to >150m. Slate, schist, quartzite, gneiss are the main hardrock types that were encountered in the wells which constitute the basement over which the alluvial material is deposited.

It is observed from pre monsoon 2010 data the ground water table varies from 170m amsl to 350m amsl.

PLATE - XXIII



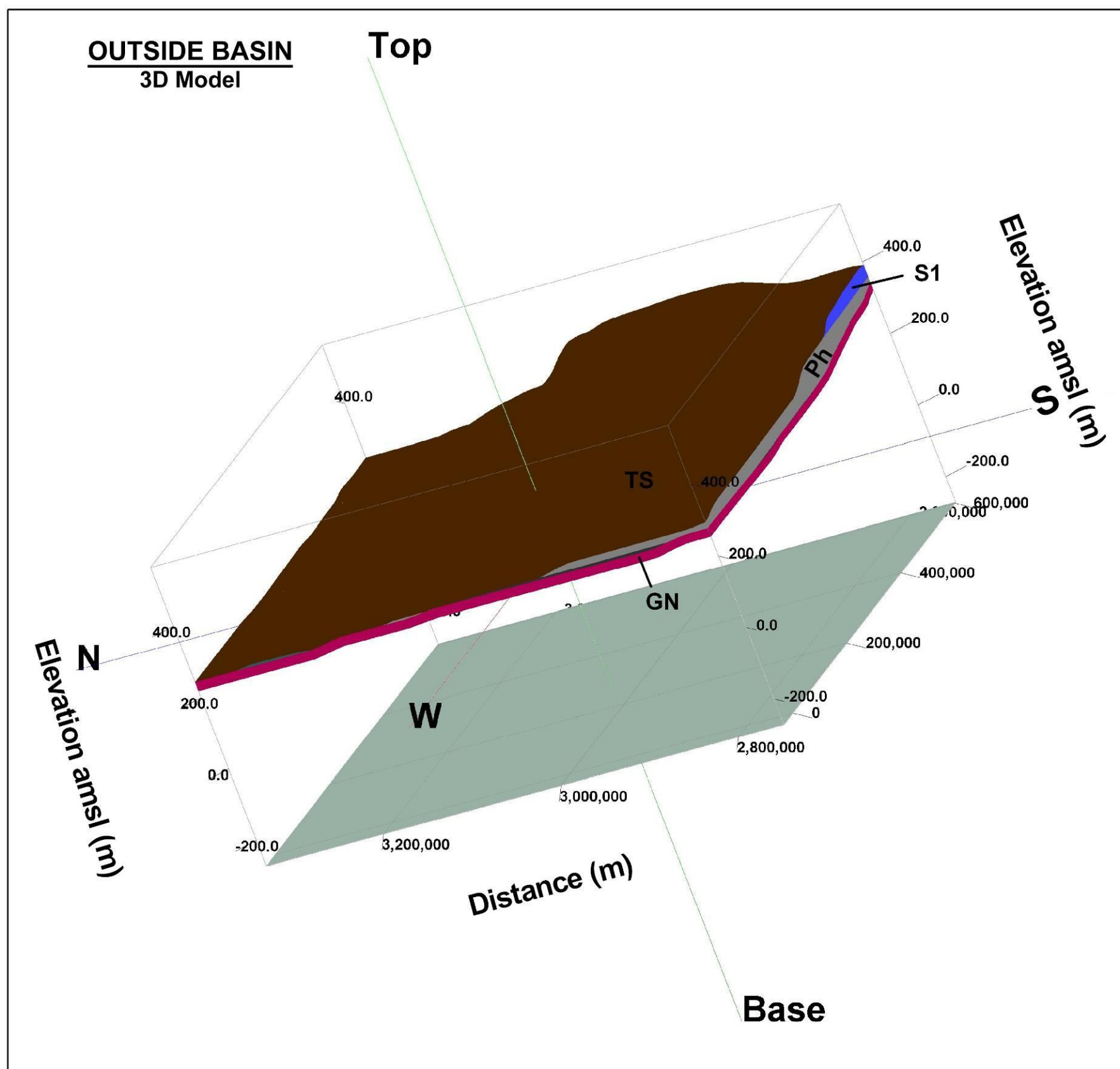
3D MODEL OF AQUIFERS

OUTSIDE BASIN

The three dimensional litho-stratigraphic model has been developed for the Other Nallahs river basin using interpolation techniques on the data of scattered wells given as an input. Plate - XXIV represents a 3D model depicting the various litho-stratigraphic units in the entire river basin. The model reveals that gneiss is a continuous formation in the area overlain by limestone and quartzite formations and in the southeast.

The depth of bedrock is low in the eastern part as compared to the western part of the basin. Granite gneiss is the basement forming rock in this basin. The small discontinuous patches depicted in 3D model are indicative of occurrence of limestone, quartzite, weathered & fractured zone as reported in data of GWD and CGWB. In absence of continuity of the similar formation in adjacent wells they appear as small patches.

PLATE - XXIV



Legend

TS	Top Soil
S1	Sand
C1	Clay
S2	Sand
C2	Clay
WFZ	Weathered & Fractured Zone
Sst	Sandstone
Lst	Limestone
Q	Quartzite
Sh	Shale
Ph	Phyllite
Sc	Schist
SL	Slate
Rh	Rhyolite
G	Granite
GN	Granite Gneiss

Glossary of terms

S. No.	Technical Terms	Definition
1	AQUIFER	A saturated geological formation which has good permeability to supply sufficient quantity of water to a Tube well, well or spring.
2	ARID CLIMATE	Climate characterized by high evaporation and low precipitation.
3	ARTIFICIAL RECHARGE	Addition of water to a ground water reservoir by man-made activity
4	CLIMATE	The sum total of all atmospheric or meteorological influences principally temperature, moisture, wind, pressure and evaporation of a region.
5	CONFINED AQUIFER	A water bearing strata having confined impermeable overburden. In this aquifer, water level represents the piezometric head.
6	CONTAMINATION	Introduction of undesirable substance, normally not found in water, which renders the water unfit for its intended use.
7	DRAWDOWN	The drawdown is the depth by which water level is lowered.
8	FRESH WATER	Water suitable for drinking purpose.
9	GROUND WATER	Water found below the land surface.
10	GROUND WATER BASIN	A hydro-geologic unit containing one large aquifer or several connected and interrelated aquifers.
11	GROUND WATER RECHARGE	The natural infiltration of surface water into the ground.
12	HARD WATER	The water which does not produce sufficient foam with soap.
13	HYDRAULIC CONDUCTIVITY	A constant that serves as a measure of permeability of porous medium.
14	HYDROGEOLOGY	The science related with the ground water.
15	HUMID CLIMATE	The area having high moisture content.
16	ISOHYET	A line of equal amount of rainfall.
17	METEOROLOGY	Science of the atmosphere.
18	PERCOLATION	It is flow through a porous substance.
19	PERMEABILITY	The property or capacity of a soil or rock for transmitting water.
20	pH	Value of hydrogen-ion concentration in water. Used as an indicator of acidity (pH < 7) or alkalinity (pH > 7).
21	PIEZOMETRIC HEAD	Elevation to which water will rise in a piezometers.
22	RECHARGE	It is a natural or artificial process by which water is added from outside to the aquifer.
23	SAFE YIELD	Amount of water which can be extracted from ground water without producing undesirable effect.
24	SALINITY	Concentration of dissolved salts.
25	SEMI-ARID	An area is considered semiarid having annual rainfall between 10-20 inches.
26	SEMI-CONFINED AQUIFER	Aquifer overlain and/or underlain by a relatively thin semi-pervious layer.
27	SPECIFIC YIELD	Quantity of water which is released by a formation after its complete saturation.
28	TOTAL DISSOLVED SOLIDS	Total weight of dissolved mineral constituents in water per unit volume (or weight) of water in the sample.

S. No.	Technical Terms	Definition
29	TRANSMISSIBILITY	It is defined as the rate of flow through an aquifer of unit width and total saturation depth under unit hydraulic gradient. It is equal to product of full saturation depth of aquifer and its coefficient of permeability.
30	UNCONFINED AQUIFER	A water bearing formation having permeable overburden. The water table forms the upper boundary of the aquifer.
31	UNSATURATED ZONE	The zone below the land surface in which pore space contains both water and air.
32	WATER CONSERVATION	Optimal use and proper storage of water.
33	WATER RESOURCES	Availability of surface and ground water.
34	WATER RESOURCES MANAGEMENT	Planned development, distribution and use of water resources.
35	WATER TABLE	Water table is the upper surface of the zone of saturation at atmospheric pressure.
36	ZONE OF SATURATION	The ground in which all pores are completely filled with water.
37	ELECTRICAL CONDUCTIVITY	Flow of free ions in the water at 25C mu/cm.
38	CROSS SECTION	A Vertical Projection showing sub-surface formations encountered in a specific plane.
39	3-D PICTURE	A structure showing all three dimensions i.e. length, width and depth.
40	GWD	Ground Water Department
41	CGWB	Central Ground Water Board
42	CGWA	Central Ground Water Authority
43	SWRPD	State Water Resources Planning Department
44	EU-SPP	European Union State Partnership Programme
45	TOPOGRAPHY	Details of drainage lines and physical features of land surface on a map.
46	GEOLOGY	The science related with the Earth.
47	GEOMORPHOLOGY	The description and interpretation of land forms.
48	PRE MONSOON SURVEY	Monitoring of Ground Water level from the selected DKW/Piezometer before Monsoon (carried out between 15th May to 15th June)
49	POST-MONSOON SURVEY	Monitoring of Ground Water level from the selected DKW/Piezometer after Monsoon (carried out between 15th October to 15th November)
50	PIEZOMETER	A non-pumping small diameter bore hole used for monitoring of static water level.
51	GROUND WATER FLUCTUATION	Change in static water level below ground level.
52	WATER TABLE	The static water level found in unconfined aquifer.
53	DEPTH OF BED ROCK	Hard & compact rock encountered below land Surface.
54	G.W. MONITORING STATION	Dug wells selected on grid basis for monitoring of state water level.
55	EOLIAN DEPOSITS	Wind-blown sand deposits

(Contd...)

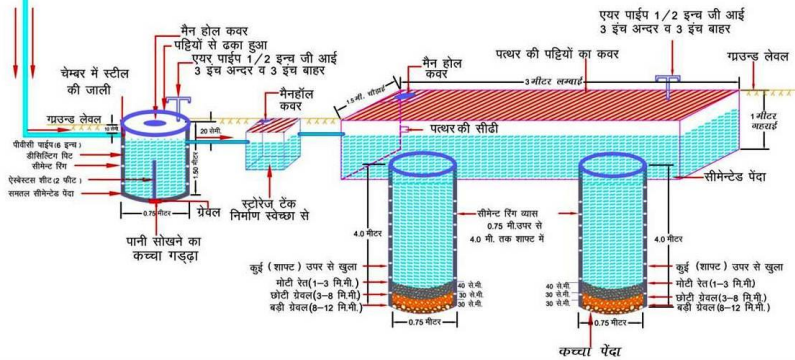


भवन छत क्षेत्रफल 300 से 500 वर्गमीटर तक

चित्र-4

निर्माण किये जाने वाले मुख्य भाग एवं डिज़ाईन

- PVC पाईप 6" व्यास
- सीमेन्ट रिंग से निर्मित डीसिल्टिंग पिट
(0.75 मी.व्यास x 1.50 मी.गहरा)
- रीचार्ज टैंक 1.5 मी. चौड़ा x 3 मी. लम्बा x 1 मी. गहरा
- सीमेन्ट रिंग से निर्मित शापट
(0.75 मी.व्यास x 4 मी.गहरा) (संख्या 2)
- संरचना की अनुमानित लागत रु 24,000 अधिक
- वार्षिक पुनर्गति जल लगभग 2,00,000 लीटर
- 20 वर्षों में पुनर्गति जल लगभग 40,00,000 लीटर
- पुनर्गति जल की लागत 1 पैसे प्रति लीटर से कम



भूजल में घुले मुख्य तत्वों की अधिकता का मानव शरीर पर दुष्प्रभाव

बोरोन-स्नायु तन्त्र पर प्रभाव

फ्लोराइड - दंत क्षरण

क्लोराइड-सोडियम के साथ
मिलकर उच्च रक्त चाप

सोडियम-हृदय, गुर्दा व रक्त
परिसंचरण रोगों से
ग्रसित लोगों को हानिकारक

कैल्शियम-जोड़ों में कड़ापन

**नाइट्रेट-नवजात शिशुओं में
ब्लू बेबी बीमारी
(मेथमोग्लोबिनिमिया)**

आर्सेनिक-त्वचा रोग, कैंसर

सल्फेट-अधिकता में मैग्नेशियम के साथ मिलकर दस्तावर

लेड-बच्चों के शारीरिक
व मानसिक विकास में बाधा
वयस्कों में गुर्दे के रोग

आयरन-आयरन जीवाणु से आमाशय संबंधी रोग

फ्लोराइड-जोड़ों में अकड़न,
हड्डियों में मुड़ाव

भूजल अमूल्य है इसे प्रदूषित न करें।

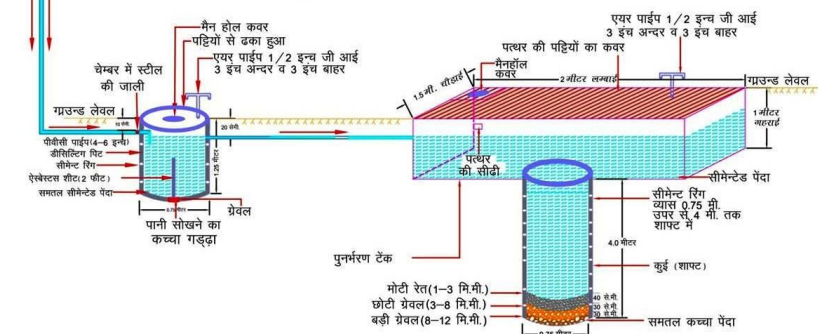


भवन छत क्षेत्रफल 200 से 300 वर्गमीटर तक

चित्र-3

निर्माण किये जाने वाले मुख्य भाग एवं डिज़ाइन

- PVC पाईप 4" – 6" व्यास
- सीमेन्ट रिंग से निर्मित डीसिल्टिंग पिट
(0.75 मी व्यास x 1.25 मी गहरा)
- शैवाज टैंक 1.5 मी. चौड़ा x 2 मी. लम्बा x 1 मी. गहरा
- सीमेन्ट रिंग से निर्मित शाफ्ट
(0.75 मी व्यास x 4 मी. गहरी)
- संरचना की अनुमानित लागत रु 15,000–16,000
- वार्षिक पुनर्भरित जल लगभग 1,25,000 लीटर
- 20 वर्षों में पुनर्भरित जल लगभग 25,00,000 लीटर
- पुनर्भरित जल की लागत 1 पैसे प्रति लीटर से कम

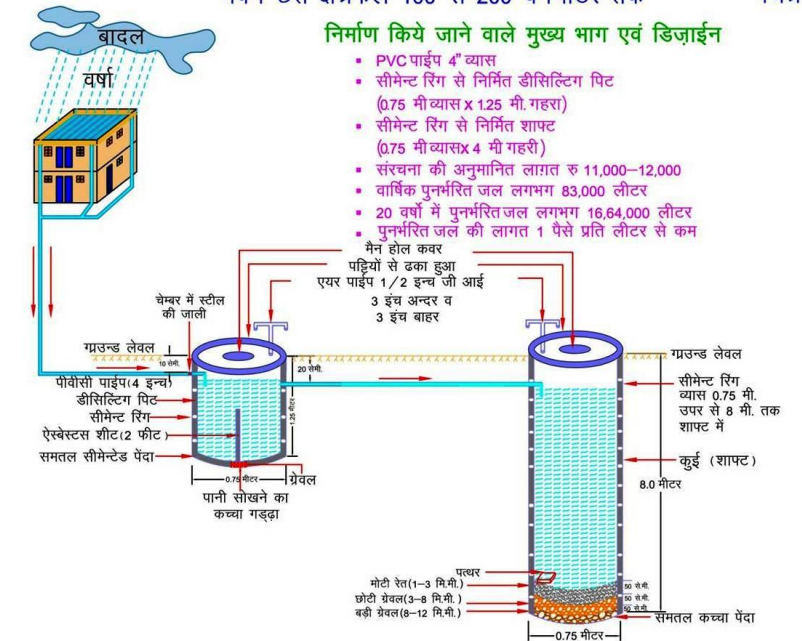


भवन छत क्षेत्रफल 100 से 200 वर्गमीटर तक

चित्र-2

निर्माण किये जाने वाले मुख्य भाग एवं डिज़ाइन

- PVC पाईप 4" व्यास
- सीमेन्ट रिंग से निर्मित डीसीलिंग पिट
(0.75 मी.व्यास X 125 मी. गहरा)
- सीमेन्ट रिंग से निर्मित शाफ्ट
(0.75 मी.व्यास X 4 मी. गहरी)
- संरचना की अनुमानित लागत रु 11,000—12,000
- वार्षिक पुनर्निर्मित जल लगभग 83,000 लीटर
- 20 वर्षों में पुनर्निर्मित जल लगभग 16,84,000 लीटर
- पुनर्निर्मित जल की लागत 1 पैसे प्रति लीटर से कम

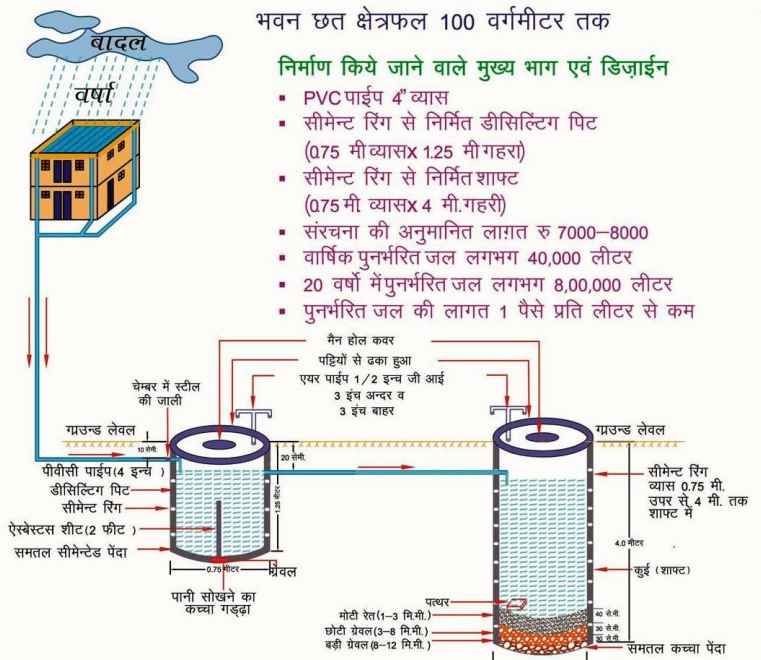


भवन छत क्षेत्रफल 100 वर्गमीटर तक

चित्र-1

निर्माण किये जाने वाले मुख्य भाग एवं डिज़ाइन

- PVC पाइप 4" व्यास
- सीमेन्ट रिंग से निर्मित डीसिल्टिंग पिट
(0.75 मी व्यास x 1.25 मी गहरा)
- सीमेन्ट रिंग से निर्मित शाफ्ट
(0.75 मी व्यास x 4 मी गहरी)
- संरचना की अनुमानित लागत रु 7000–8000
- वार्षिक पुनर्भरित जल लगभग 40,000 लीटर
- 20 वर्षों में पुनर्भरित जल लगभग 8,00,000 लीटर
- पुनर्भरित जल की लागत 1 पैसे प्रति लीटर से कम



Myths and Facts about Ground Water

S No	Myths	Facts
1	What is Ground Water • an underground lake • a net work of underground rivers • a bowl filled with water	Water which occurs below the land in geological formations/rocks is Ground water
2	Ground Water occurs everywhere beneath the Land Surface	Not really, it depends on the nature of rock formation
3	There is a relationship between ground water and surface water	Not all the places. Near streams/rivers there is relation
4	Groundwater is not renewable resource	It is renewable source and every year it is being recharged through rain/applied irrigation etc
5	Ground water is unlimited and deeper you drill more discharge	It is limited to annual recharge from rain/applied irrigation. The discharge may not increase if you go deeper
6	Ground Water moves rapidly	The movement of ground water is very slow
7	Ground water pumped from wells is thousands of years old	Generally the ground water being tapped through wells is a few years old
8	If water taste good—it is safe to drink	It may have other chemicals e.g. fluoride, nitrates etc which are harmful
9	Water from free flowing tube wells is very pure	This water can also be contaminated so test before use
10	If I recharge my TW/DW/HP it will not benefit me	It will also benefit you and also adjoining wells
11	There is no static ground water resources in Rajasthan	Rajasthan is also having Static GW resources, and being tapped in most of areas as GW annual withdrawal is more than annual recharge
12	I cannot meet annual cooking and drinking water requirement by rain water harvesting	The water requirement for drinking and cooking is only 8 lit/day. You can harvest this water for family of 5 persons from roof top or paved area of 75 Sq m to meet annual requirement
13	You can increase ground water recharge	This can be done by harvesting the rain water and storing in sub surface reservoir (GW) by constructing the recharge structures
14	You cannot use abandoned TW/HP/DW for ground water recharge	These should be used as recharge structures as harvested rain water is directly put into GW reservoir
15	Putting waste near HP/TW will not cause any problem	Such actions will pollute wells and water



ROLTA

Rolta India Limited

Central & Registered Office

Rolta Tower A,

Rolta Technology Park,

MIDC, Andheri (East),

Mumbai - 400 093

Tel : +91 (22) 2926 6666, 3087 6543

Fax : +91 (22) 2836 5992

Email : indsales@rolta.com

www.rolta.com